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Review

A review on MPS method developments and applications in nuclear engineering

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Abstract

The paper aims to provide a comprehensive review on the achievements made on moving particle semi-implicit (MPS) method and the applications in nuclear engineering. The state-of-the-art advancements in stability and accuracy, boundary conditions, surface tension, multiphase flow, fluid–structure interaction and multi-resolution technique are discussed. The applications in nuclear reactor fundamental thermal hydraulics and melt behaviors in severe accident are summarized. The drawbacks, current challenges, future developments and potential application areas of MPS method are highlighted.

Keywords: MPS method; Stability; Accuracy; Boundary condition; Surface tension; Multiphase flow; Fluid–structure interaction; Multi-resolution; Thermal hydraulics; Severe accident

Contents

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1. Introduction

Particle methods simulate fluid flow by discretizing the computation domain into particles that carry the information of physical property, velocity, and pressure. Particle motion is implemented by solving particle interaction models in the Lagrangian framework. Particle methods have advantages in capturing fluid surfaces and calculating fluid deformation, breakup, and coalescence. At present, two prevailing particle methods are the smooth particle hydrodynamics (SPH) method [1,2] and the moving particle semi-implicit (MPS) method [3]. SPH is a method based on the integral representation of quantities and spatial derivatives, which is originally developed using a fully explicit algorithm for compressible non-viscous flows. It gets wide application in astrophysics where the density changes so much ranging from the inside of stars to vacuum, but the method application is later expanded to the solid and fluid mechanics as well. By contrast, MPS method is based on Taylor series expansion, which is originally developed for incompressible fluid flow using a semi-implicit algorithm. To date, however, as many advancements have been made to particle methods, SPH and MPS methods have similar features in fluid mechanics. For example, a pressure Poisson equation (PPE) that is solved using implicit algorithm is introduced to SPH method [4], and a fully explicit algorithm is also applied to MPS method by replacing PPE with the equation of state [5–7]. They can be both used to the simulations of compressible and incompressible flows, respectively, with the relevant improvements, such as ISPH [4,8–10] and WC-MPS [5,7]. Nevertheless, the spatial discretization formulations of the SPH and MPS methods have significant differences.

MPS method applies a prediction–correction semi-implicit computation algorithm, which calculates external force and viscous term in the prediction step and calculates pressure term in correction step. The PPE is solved implicitly to calculate pressure, and the flow field, velocity, and pressure profiles are obtained by tracking particle motion. Free surface is detected by the deficiency of particle number density. In MPS method, the neighboring particles are memorized in a neighbor list and it is updated in every time step. This is an additional procedure in each time step and time consuming in contrast to mesh generation that is carried out once in the initialization process. Since MPS method has been brought forward for only two decades, it is considered as one promising but immature method. Some drawbacks still exist with MPS method, such as low accuracy, instability and high computation cost. Many researchers have been working on the method development to improve its accuracy, stability, and the capability to simulate complex multiphase flow and fluid–structure interactions.

Nuclear engineering is the first application area of MPS method. The phenomena are related to gas–liquid two-phase flow, solid–liquid two-phase flow, phase transition, fluid–structure interaction and the flows of large deformation, breakup and coalescence. MPS method can simulate some phenomena of nuclear engineering with its inherent merits, but some additional physical models need to be developed according to the specific

Nomenclature

$\mathbf{a}$	A gradient vector
C	Color function value or a coefficient
$\mathbf{C}$	Corrective matrix
C_{p}	Specific heat capacity
d	Number of spatial dimensions
$\boldsymbol{F}$	Force
$\mathbf{g}$	Gravity
G	Gaussian function
k	Conductivity
l_0	Initial particle distance
$\mathbf{L}$	A matrix
m	Mass
n	Particle number density
$\mathbf{n}$	A normalized vector
n^0	Standard value of particle number density
P	Pressure
Q	Heat source
r_e	Cutoff radius
$\mathbf{r}$	Particle position vector
r	Distance
t	Time
T	Temperature
u	Horizontal velocity
$\mathbf{u}$	Velocity vector
v	Vertical velocity
$w(\mathrm{x})$	Weight function
x	Position

Greek symbols

β, γ	Coefficients
$\boldsymbol{\gamma}$	Surface tension force
δ	Dirac function
Δ	Increment
η	Weight coefficient
θ	Angle
κ	Curvature
λ	Coefficient of Laplacian model
μ	Dynamic viscosity

phenomena. With the efforts of researchers, a number of simple fundamental thermal hydraulic phenomena have been investigated, and in recent years, MPS method also gets application in the simulations of melt behaviors in nuclear reactor severe accident. As mentioned above, nevertheless, MPS method has the drawbacks of low accuracy and being liable to instability, which are great challenges in the practical applications. It is necessary to improve MPS method and develop numerical models aiming at the practical problems. Therefore, a comprehensive review on the current knowledge on MPS method development and applications in nuclear engineering and the challenges in modeling complex thermal hydraulic phenomena is of crucial importance for future studies.

Π	Adjusting parameter
ρ	Fluid density
σ	Surface tension coefficient
ϕ	A scalar quantity
$\boldsymbol{\varphi}$	A vector

Subscripts

G	Gaussian function
i, j	Particle identification number

Supscripts

DS	Dynamic stabilization
k	Time step number
*	Temporal value

Abbreviations

BWR	Boiling water reactor
CCI	Corium concrete interaction
CSF	Continuum surface force
DEM	Discrete element method
FCM	First-order corrective matrix
FEM	Finite element method
FSI	Fluid–structure interactions
ISPH	Incompressible SPH
LSMPS	Least squares MPS
MCCI	Molten corium-concrete interaction
MLE	Mixed-Lagrangian–Eulerian
MPS	Moving particle semi-implicit
MPS-FE	MPS-finite element
MPS-HS	MPS-high order source term
MPS-MAFL	MPS-meshless advection using flow-directional local grid
MMPS-CA	Multiphase MPS with continuous acceleration
MMPS-HD	Multiphase MPS with harmonic density
OPS	Optimized PS
PPE	Pressure Poisson equation
PS	Particle shifting
PW	Polygon wall
PWR	Pressurized water reactor
RPV	Reactor pressure vessel
SCM	Second-order corrective matrix
SPH	Smooth particle hydrodynamics
VOF	Volume of fluid
WC-MPS	Weakly compressible MPS

Review articles on particle simulation technologies for fluid dynamics have been published by Koshizuka [11] and Gotoh and Khayyer [12]. In [11], the contents mainly focus on the application areas of MPS method, while the generic improvements on MPS models are not mentioned. In [12], by contrast, special attentions are given to the technologies of particle methods, including both SPH and MPS methods, and applications in ocean engineering. At present, however, it lacks a review to summarize the developments of MPS method itself and applications in nuclear engineering to provide a reference for MPS developers and users. Therefore, the present work aims to focus on the state-of-the-art knowledge of MPS method and applications in nuclear engineering. The paper is organized as follows. At first, the technologies corresponding to MPS stability and accuracy, boundary conditions, surface tension, multiphase flow, fluid–structure interaction and multi-resolution technique are reviewed. Then, the applications in nuclear reactor fundamental thermal hydraulics and melt behavior are summarized. Finally, the discussions on the possibilities and challenges of future improvement and further potential application in nuclear engineering complete the paper.

2. MPS method development

The mass, momentum and energy conservation equations are the governing equations.

$$\frac{1}{\rho}\frac{D\rho}{Dt} = \nabla \cdot \mathbf{u} = 0 \tag{1}$$

$$\frac{D\mathbf{u}}{Dt} = -\frac{1}{\rho}\nabla P + \frac{\mu}{\rho}\nabla^2\mathbf{u} + \mathbf{g} \tag{2}$$

$$\rho C_p \frac{DT}{Dt} = k\nabla^2 T + Q \tag{3}$$

where $\mathbf{u}$ is velocity, P is pressure, $\mathbf{g}$ is gravity, T is temperature, ρ is density, C_p is specific heat capacity, μ is dynamic viscosity, k is conductivity, and Q is heat source. Mass conservation equation (Eq. (1)) is written in the form of incompressible flow. Particle number density is proportional to the fluid density and it is sustained constant at each time step to satisfy fluid incompressibility.

In MPS method, the particle motion is implemented through calculating particle interaction models for every particle with its neighboring particles. The contributions from neighboring particles to the target particle are evaluated by weight function. The differential operators in the governing equations, such a gradient, Laplacian, and divergence, are usually discretized with the following equations [3].

$$\langle \nabla \phi \rangle_i = \frac{d}{n^0}\sum_{j\neq i}\frac{\phi_j - \phi_i}{|\mathbf{r}_j - \mathbf{r}_i|^2}(\mathbf{r}_j - \mathbf{r}_i)w(|\mathbf{r}_j - \mathbf{r}_i|) \tag{4}$$

$$\left\langle \nabla^2 \phi \right\rangle_i = \frac{2d}{\lambda n^0}\sum_{j\neq i}(\phi_j - \phi_i)w(|\mathbf{r}_j - \mathbf{r}_i|) \tag{5}$$

$$\langle \nabla \cdot \boldsymbol{\varphi} \rangle_i = \frac{d}{n^0}\sum_{j\neq i}\frac{(\boldsymbol{\varphi}_j - \boldsymbol{\varphi}_i)(\mathbf{r}_j - \mathbf{r}_i)}{|\mathbf{r}_j - \mathbf{r}_i|^2}w(|\mathbf{r}_j - \mathbf{r}_i|) \tag{6}$$

where ϕ and $\boldsymbol{\varphi}$ represent a vector and scalar, respectively, $w(\mathrm{r})$ is weight function, n^0 is the standard value of particle number density, λ is the coefficient of Laplacian model, d is the number of spatial dimensions, $\mathbf{r}$ is particle position vector, and subscripts i and j denote particle i and j.

2.1. Stability and accuracy

Stability and accuracy are closely related to each other, which are often improved simultaneously in past studies [13–15]. Therefore, we discuss herein the improvement techniques of stability and accuracy together. Unphysical pressure oscillations and instabilities are inherent drawbacks of MPS method, which link with the non-uniformity of particle distribution. Perturbations in particle motion will lead to maldistribution of particles and such perturbations will also lead to spatial discontinuities in the calculation of PPE, and accordingly result in inaccuracies in the calculated pressure field. In the absence of stabilization or particle regularization schemes, such growth of perturbations in particle motion will eventually lead to particle clumping or splashing and even break-up of calculation.

2.1.1. Gradient model with stabilizing force

Stress instability is an inherent drawback of particle method, affecting the accuracy and convergence of numerical simulations. It can be classified as tensile instability that occurs in presence of attractive inter-particle forces when particles are disordered or clustering, and compressive instability that occurs in presence of repulsive inter-particle forces when particles are under compressive stress states. Khayyer and Gotoh [13] investigated the mechanism of tensile instability and pointed out that MPS-based simulations are prone to become destabilized in presence of attractive inter-particle forces. To solve such instability problem, the concept of repulsive inter-particle forces for stability enhancement is proposed by Koshizuka et al. [16] by modifying the pressure gradient model, in which the pressure of the target particle is replaced by the minimum pressure among the neighboring particles, so that the pressure interacting forces will be purely repulsive. This approach applies an artificially repulsive force to drive particles from dense to sparse regions, and the forces are balanced for perfectly symmetrical distribution of neighboring particles. Some congeneric modifications are brought forward based on this repulsive force concept. For example, Khayyer and Gotoh [17] proposed a new pressure gradient formulation based on the nearly exact conservation of linear and angular momentums in MPS calculation.

$$\langle \nabla P \rangle_i = \frac{d}{n_0} \sum_{i \neq j} \frac{(P_i + P_j) - (\hat{P}_i + \hat{P}_j)}{\left|\mathbf{r}_j - \mathbf{r}_i\right|^2} (\mathbf{r}_j - \mathbf{r}_i) w(\left|\mathbf{r}_j - \mathbf{r}_i\right|) \tag{7}$$

Tsuruta et al. [18] pointed out that the stability schemes in Koshuzika's and Gotoh's formulations depend on pressure states and particle distributions, and the artificial repulsive force is predominant rather than the original gradient force. As a result, particles may present unphysical motions and perturbations because of overestimation of inter-particle pressure forces. Hence, a dynamically stabilized gradient operator is proposed by incorporating a meticulously adequate stabilizing force to the original gradient model.

$$\langle \nabla P \rangle_i = \frac{d}{n_0} \sum_{j \neq i} \frac{(P_j - P_i)}{\left|\mathbf{r}_j - \mathbf{r}_i\right|^2} (\mathbf{r}_j - \mathbf{r}_i) w(\left|\mathbf{r}_j - \mathbf{r}_i\right|) + \frac{1}{n_0} \sum_{i \neq j} \mathbf{F}_{ij}^{DS} w(\left|\mathbf{r}_j - \mathbf{r}_i\right|) \tag{8}$$

$$\begin{cases} \mathbf{F}_{ij}^{DS} = 0 & if \quad \left|\mathbf{r}_{ij}^*\right| \geq d_{ij} \\ \mathbf{F}_{ij}^{DS} = -\rho_i \prod_{ij} \mathbf{e}_{ij} & if \quad \left|\mathbf{r}_{ij}^*\right| < d_{ij} \end{cases} \tag{9}$$

where $\mathbf{F}_{ij}^{DS}$ is stabilizing force from neighboring particle j to target particle i, Π_{ij} is a parameter to adjust the magnitude of $\mathbf{F}_{ij}^{DS}$. Xiang and Chen [19] proposed an inter-particle force stabilization and consistency restoring model to overcome the compressive instability. A hyperbolic-shaped quintic kernel function is developed with a non-negative and smooth second order derivative for inter-particle force stabilization; a Taylor series expansion is also derived for consistency restoring. Liu et al. [20] derived a high-order stabilized pressure gradient model to achieve purely repulsive inter-particle force and the first-order Taylor series consistency, with the assistance of using dummy neighboring particles. Jandaghian and Shakibaeinia [7] proposed a new conservative gradient model with a dynamic particle stabilizing term that takes into account the particle number density and pressure values at particles i and j.

2.1.2. Improvement of pressure Poisson equations

The formulation of PPE source term is an important cause of pressure oscillation and density error accumulation. Ikeda et al. [21] proposed to improve the stability by introducing artificial compressibility, but the coefficient of compressibility depends on the spatial resolution and time step, and it will also bring an adverse effect to numerical convergence. Tanaka and Masunaga [22] developed an artificially compressible model that has no relation with time step, but the effect on the convergence cannot be completely eliminated. Khayyer and Gotoh [23] proposed a new artificially compressible model based on the pressure distribution at previous time step, which effectively suppresses the pressure oscillation. On the other aspect, some researchers [22–24] applied mixed source term consisting of the divergence of temporal velocity and the relative variation of particle number density; the former can make pressure be smooth in terms of both space and time, while the later can preserve incompressibility. Combined with Ikeda's artificially compressible model, the new PPE is written as

$$\langle \nabla^2 P \rangle_i^{k+1} = (1-\gamma) \frac{\rho_0}{\Delta t} \nabla \cdot \mathbf{u}^* - \gamma \frac{\rho_0}{\Delta t^2} \left(\frac{n^* - n^0}{n^0} \right) + \alpha \frac{\rho_0}{\Delta t^2} P_i^{k+1} \tag{10}$$

where γ is an adjusting coefficient from 0.01 to 0.05, and α is an artificially compressible coefficient, which is usually set at 10^{-9}-10^{-6}.

To improve the accuracy of pressure calculation, Khayyer and Gotoh [23] derived a high-order formulation for PPE source term based on the time differentiation of particle number density that is calculated using the standard MPS kernel function. The modified MPS method is named as MPS-HS.

$$\left\langle \nabla^2 P_{k+1} \right\rangle_i = \frac{\rho}{n_0 \Delta t}\left(\frac{Dn}{Dt}\right)^* = -\frac{\rho}{n_0 \Delta t}\left(\sum_{j\neq i}\frac{r_e}{r_{ij}^3}(x_{ij}u_{ij} + y_{ij}v_{ij})\right)^* \tag{11}$$

Even though more stable and accurate pressure calculation is achieved by implementing the high-order source term, together with stabilization technique and high-order Laplacian model, the unphysical numerical oscillations still exist in the studies of violent sloshing flow. Kondo and Koshizuka [25] derived a mixed formulation for PPE source term consisting of one main part and two error-compensating parts, as shown in Eq. (12). By adjusting the coefficients (β and γ) of two error-compensating parts, smooth pressure distribution can be attained. A disadvantage of this technique is that the optimal coefficient pair is related to the time step and particle spatial resolution. Thus, they need to be calibrated in the practical application.

$$\frac{1}{\rho^0}\left(\frac{D^2\rho}{Dt^2}\right)_i = \frac{n_i^* - 2n_i^k + n_i^{k-1}}{n^0 \Delta t^2} + \frac{\beta}{\Delta t}\frac{n_i^k - n_i^{k-1}}{n^0 \Delta t} + \frac{\gamma}{\Delta t^2}\frac{n_i^k - n^0}{n^0} \tag{12}$$

On the basis of Kondo and Koshizuka' model, Khayyer and Gotoh [13] dynamically calculated the adjusting coefficients of error-compensating parts, which are the functions of instantaneous flow.

In term of Laplacian model in PPE, a diffusion coefficient λ is used in the original model to ensure the increase in variance be equivalent to that of the analytical solution, but this diffusion coefficient may cause numerical difficulties in calculating PPE. Zhang et al. [26] and Xu and Jin [14] derived new Laplacian models by the divergence of gradient without using unphysical coefficient. Duan and Chen [27] also derived a new Laplacian model by using Gaussian function as the weight function. On the other hand, Isshiki [28] visited the theoretical background of differential operators of MPS method, and applied the Gauss divergence theorem on a circular control volume to reproduce MPS models. Inspired by Isshiki's work, Ng et al. [29] derived a more general Laplacian model on both the regular and irregular node structures. The accuracy of several Laplacian models that are produced by changing circular radii and particle numbers at circular boundary is evaluated in solving PPE under both Dirichlet and Neumann boundary conditions.

In regards to the high-order formulation of Laplacian model, Khayyer and Gotoh [30] derived the following equation

$$\begin{aligned}\nabla \cdot \langle \nabla \phi \rangle_i &= \frac{1}{n_0}\sum_{j\neq i}\left(\frac{2\phi_{ji}}{r_{ij}}\frac{\partial w_{ij}}{\partial r_{ij}} + \phi_{ji}\frac{\partial^2 w_{ij}}{\partial r_{ij}^2} + \frac{\phi_{ij}}{r_{ij}}\frac{\partial w_{ij}}{\partial r_{ij}}\right) \\ &= \frac{1}{n_0}\sum_{j\neq i}\left(\phi_{ji}\frac{\partial^2 w_{ij}}{\partial r_{ij}^2} - \frac{\phi_{ji}}{r_{ij}}\frac{\partial w_{ij}}{\partial r_{ij}}\right)\end{aligned} \tag{13}$$

This high-order Laplacian model was further expanded to 3D in [31]. Tamai et al. [32] conducted a study on the consistency and convergence for the Laplacian operator and proposed a novel discretization scheme for the second order differential operator, which is the basis function for least squares MPS method discussed in Section 2.1.5. In addition, Hirata and Anzai [33] proposed to stabilize the calculation of slow fluid flow by using a multiple relaxation method, which calculates the PPE more than once and the particle velocity and pressure are updated several times accordingly.

2.1.3. Corrective matrix for particle interaction models

A corrective matrix (Eqs. (14) and (15)) derived from Taylor series expansion is first applied to pressure gradient model by Khayyer and Gotoh [13] to enable a more symmetrical particle distribution and to enhance the computation accuracy. The corrective matrix C_{ij} is approximate to be a unit matrix when particle distribution is uniform. However, the corrective matrix has no contribution to the computation stabilities.

$$\langle \nabla P \rangle_i = \frac{D_s}{n_0}\sum_{j\neq i}\frac{P_j - P_i}{\left|\mathbf{r}_j - \mathbf{r}_i\right|^2}(\mathbf{r}_j - \mathbf{r}_i)\mathbf{C}_{ij}\, w(\left|\mathbf{r}_j - \mathbf{r}_i\right|) \tag{14}$$

$$\mathbf{C}_{ij} = \begin{pmatrix} \sum V_{ij} \dfrac{w_{ij} x_{ij}^2}{r_{ij}^2} & \sum V_{ij} \dfrac{w_{ij} x_{ij} y_{ij}}{r_{ij}^2} \\ \sum V_{ij} \dfrac{w_{ij} x_{ij} y_{ij}}{r_{ij}^2} & \sum V_{ij} \dfrac{w_{ij} y_{ij}^2}{r_{ij}^2} \end{pmatrix}^{-1} \tag{15}$$

Duan et al. [15] extended the application of corrective matrix to all the particle interaction models, including gradient, divergence and Laplacian models, to improve the accuracy comprehensively. The modified gradient model is the same as that of Khayyer and Gotoh's formulation, while the divergence and Laplacian models are written as Eqs. (16) and (17).

$$\langle \nabla \cdot \mathbf{u} \rangle_i = \frac{d}{n^0} \sum_{j \neq i} \left\{ \frac{\mathbf{u}_j - \mathbf{u}_i}{|\mathbf{r}_j - \mathbf{r}_i|} \cdot (\mathbf{C}_i \frac{\mathbf{r}_j - \mathbf{r}_i}{|\mathbf{r}_j - \mathbf{r}_i|}) w_{ij} \right\} \tag{16}$$

$$\left\langle \nabla^2 \phi \right\rangle_i = \frac{d}{n_0} \sum_{j \neq i} \left\{ (\phi_j - \phi_i) \left(\frac{2}{\lambda} - \frac{\mathbf{L}_i \mathbf{C}_i (\mathbf{r}_j - \mathbf{r}_i)}{|\mathbf{r}_j - \mathbf{r}_i|^2} \right) w_{ij} \right\} \tag{17}$$

where $\mathbf{L}_i \mathbf{C}_i (\mathbf{r}_j - \mathbf{r}_i)$ is a dimensionless scalar. Duan et al. [34] conducted a comprehensive analysis on the truncation and stabilization errors to identify which one is the dominant error. The use of corrective matrix converts the large random discretization error that is caused by disordered particle distribution to the high-order truncation error. Whereas, the stabilization error comes from some particle adjustment strategies for stability. The classification of errors in MPS method is shown in Fig. 1. The changes of total pressure errors when using the first order and second order corrective matrix (FCM and SCM) to all the high-order particle interaction models are evaluated. The comparison indicates that the total error reduction is insignificant when using the SCM scheme, as shown in Fig. 2, which means that truncation error is not the dominant error compared to stabilization error. The second order corrective matrix is written as

$$\mathbf{C} = \begin{bmatrix} (P_1, P_1) & (P_1, P_2) & \cdots & (P_1, P_5) \\ (P_2, P_1) & (P_2, P_2) & \cdots & (P_2, P_5) \\ \vdots & \vdots & \ddots & \vdots \\ (P_5, P_1) & (P_5, P_2) & \ldots & (P_5, P_5) \end{bmatrix}^{-1} \tag{18}$$

where $(P_\alpha, P_\beta) = \sum_{j \neq i} P_\alpha \cdot P_\beta \frac{w_{ij}}{n_0}$, and

$$\mathbf{P} = \begin{bmatrix} \dfrac{x_{ij}}{r_{ij}} & \dfrac{y_{ij}}{r_{ij}} & \dfrac{x_{ij}^2}{l_0 r_{ij}} & \dfrac{y_{ij}^2}{l_0 r_{ij}} & \dfrac{x_{ij} y_{ij}}{l_0 r_{ij}} \end{bmatrix}^T = \begin{bmatrix} P_1 & P_2 & P_3 & P_4 & P_5 \end{bmatrix}^T$$

Liu et al. [20] derived high-order stabilized gradient model, Laplacian model and divergence model by introducing dummy neighboring particles to produce error-compensating terms. Corrective matrixes are introduced to the discretization processes of differential operators using Taylor series expansions, and an accurate source term of PPE is developed by applying corrective matrix to the divergence model.

2.1.4. Particle shifting scheme

Particle shifting (PS) scheme is one of particle regularization techniques based on the concept of Fick's diffusion law, which drives particle motion from high concentration to low concentration regions. With this artificial treatment, particle distribution can be maintained in a nearly uniform pattern, and then differential operators are calculated more correctly. PS scheme is first proposed by Xu et al. [8] in SPH method and then extended to a generalized form by Lind et al. [35]. Aiming at the challenges encountered when applying PS scheme to free surface and interface of multiphase flow, where the shifting normal to the interface should be eliminated, Khayyer et al. [36] proposed an optimized PS (OPS) scheme to implement only the tangential shifting for free surface particles. In contrast to the original PS scheme, the OPS scheme is free of any adjusting parameters, and provides regular particle distributions and the results are more accurate. Khayyer et al. [37] also applied OPS scheme to the simulation of multiphase flow. The implementation process is divided into two steps: at the first step, shifting of heavy phase particles is conducted by ignoring light phase particles; then, both heavy and light phase particles are considered in the second

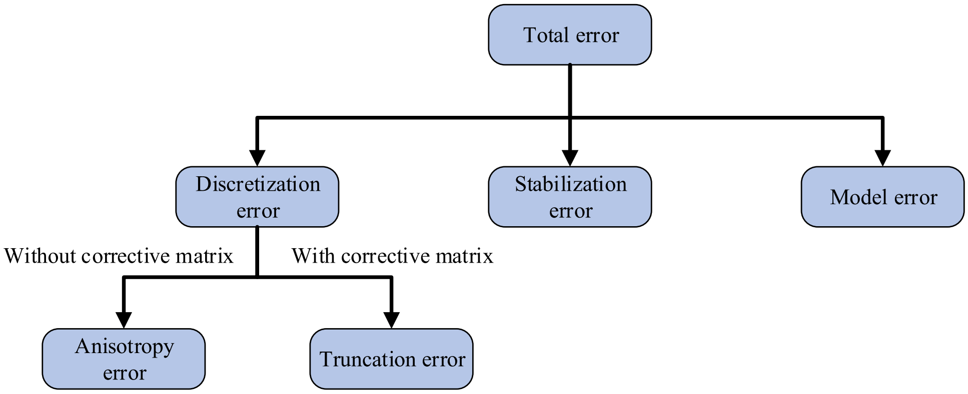


Fig. 1. The error classification in the MPS method [34].

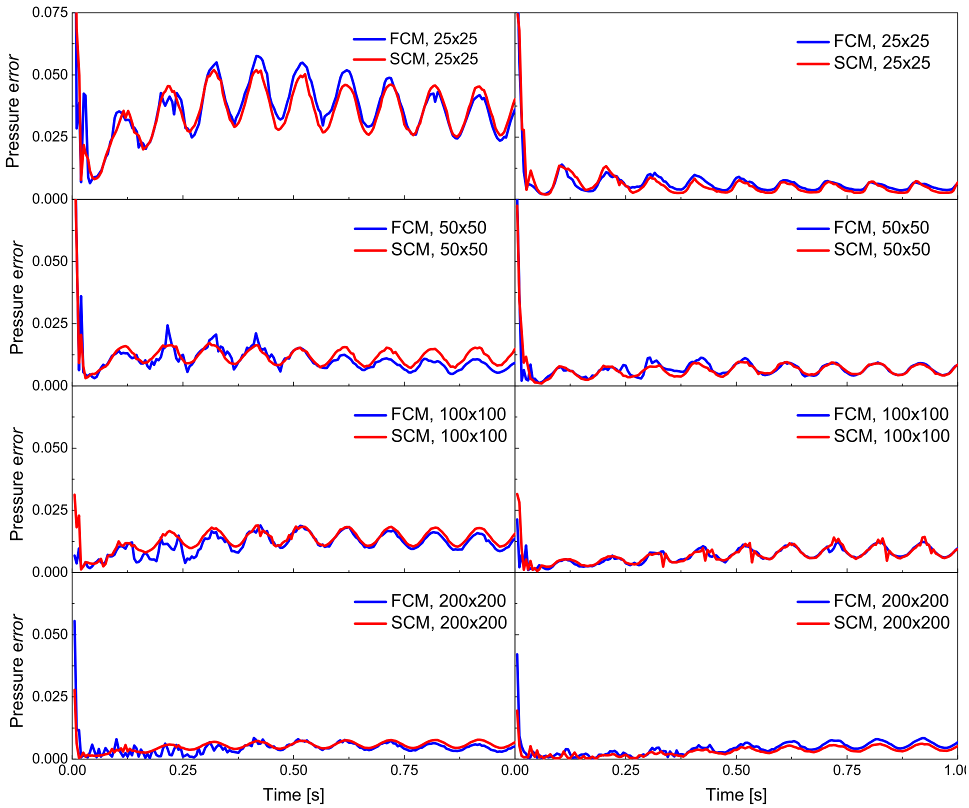


Fig. 2. Comparison of pressure dimensionless root-mean-square error simulated by the FCM and SCM schemes [34].

shifting process. Particle penetration between two different phases is effectively avoided. Duan et al. [15] introduced a non-dimensional correction matrix to PS scheme to stabilize pressure calculation. Jandaghian and Shakibaeinia [7] improved Lind's PS scheme by dividing fluid particles into four groups: free surface, free surface vicinity, internal and external or splashed particles, to apply different local shifting vectors for every region. The purpose of applying such detailed process is to vanish particle diffusion at normal direction to the free surface. The improved PS scheme is combined with pair-wise particle collision model as well to avoid particle clustering and associated noises. On the other hand, Xu and Jin [14] developed a particle velocity correction technique based on the continuity equation to suppress pressure noise and unphysical pressure fluctuations. Xu and Jin [38] calculated the corrections

Table 1
Discretization schemes and weight functions adopted for differential operators in Matsunaga et al. [41].

	Derivatives	Scheme	Weight function
Fluid particle	$\nabla^2 \mathbf{u}^k$	Type-A	Standard
	$\delta \mathbf{u}^k$	Type-A	Upwind
	$\nabla \cdot \mathbf{u}*$	Type-B	Standard
	$\nabla^2 p^{k+1}$	Type-A	Standard
	∇p^{k+1}	Type-A	Standard
Surface node	$\nabla^2 \mathbf{u}^k$	Type-A with constraint	Standard
	$\delta \mathbf{u}^k$	Type-A with constraint	Upwind
	$\nabla \otimes \mathbf{u}*$	Type-B	Standard
	$(\nabla \otimes \nabla) p^{k+1}$	Type-A	Standard
	∇p^{k+1}	Type-A	Inward

of particle positions based on the Leibniz–Reynolds transport theorem that first calculates the volume variation for every particle.

2.1.5. Least squares MPS method

Tamai and Koshizuka [39] developed least squares MPS (LSMPS) method, which adopts arbitrary high-order consistent spatial discretization scheme, consistent time integration scheme, and generalized treatments of boundary conditions. LSMPS method applies the procedure of weighted "Least Squares", and follows the fundamentals of MPS method, but the formulations are very different with that of present MPS method. Remarkable enhancements in stability and accuracy and high-order consistency are demonstrated for arbitrary dimensions. It should be noted that the LSMPS method does not need to use any pressure stabilization techniques mentioned above. Hu et al. [40] introduced arbitrary Lagrangian–Eulerian technique to the LSMPS method by using an upwind interpolation scheme, and proposed spring model and static model to enhance the accuracy and numerical stability. LSMPS method has two types of formulations (type-A and type-B) to discretize the spatial derivatives. In Matsunaga et al. [41], type-B scheme is applied to the divergence of provisional velocity in PPE. With this treatment, the decoupling problem of pressure and velocity that is caused by the collocated variable arrangement is avoided. Another characteristic is that the free surface is constructed by mesh nodes, which are coupled with the calculation of fluid particles. Anisotropic weight functions are introduced for convection term (for particles/nodes) and pressure gradient term (for nodes only). Specific discretization schemes and weight functions applied in the differential operators are illustrated in Table 1. In addition, in order to avoid the accumulation of calculation error in the long-time calculations, internal fluid particles are shifted slightly at each time step. Using these treatments in several verification tests, the first order accuracies in global models including gradient, Laplacian and divergence are achieved.

2.1.6. Application of background mesh

Apart from aforementioned stability and accuracy improvement approaches, some researchers have suggested solving particle maldistribution problem by making the use of background meshes. Such scheme has many applications in the stability improvement of SPH method. In MPS method, Wang et al. [42] proposed background mesh scheme to provide an accurate, smooth and spatially continuous source term for the PPE. Source term is firstly calculated at mesh nodes with the information of interpolated velocities at fixed and regular neighboring nodes. Then, the solved source term at mesh modes is extrapolated to the neighboring particles. The imaginary background meshes are only used for PPE calculation, while the calculations of other Navier–Stokes equation terms are implemented in the Lagrangian framework without using any Eulerian formulations. Thus, this background mesh scheme keeps the advantages of Lagrangian mesh-free methods and enhances the stability and accuracy of pressure calculations. Liu et al. [43] proposed a mixed-Lagrangian–Eulerian (MLE) method based on high-order particle method and embedded pressure mesh. The spatial derivative terms in governing equations are solved on the Cartesian grids to attain higher-order accuracy and the results are interpolated to Lagrangian particles. Meanwhile, the advection term is realized by using MPS method to get rid of convective instability problem. The results of high stability and accuracy are attained with the contribution of background mesh. Hwang [44] and Ng et al.

[45] proposed an accurate MPS method by inserting pressure meshes to particle cloud to handle the continuity constraint. The pressure is calculated on grids under Eulerian framework and the computational particles are no longer treated as material points, but serve as observation points. The viscous term is solved using an implicit and consistent Laplacian model in particles cloud. With this hybrid method, strict conservation of mass flux and numerical consistency are unconditionally satisfied.

2.1.7. Brief summary

Some approaches have been proposed for the stability and accuracy of numerical simulations, such as the concept of repulsive inter-particle force, using error-compensating terms to PPE source term, high-order gradient and Laplacian models, corrective matrixes to gradient and Laplacian models, and PS scheme. However, calibration coefficients are often adopted in these models and some models may deteriorate the consistency, convergence and conservation properties of the numerical method. The overall accuracy is about one order of magnitude lower than that of mesh-based method and unphysical pressure fluctuation is not completely resolved. Instead of using above stabilizing operations, the least squares MPS method revisits the governing equations of fluid mechanics, and derives a new set of formulations of differential operators by Taylor series expansion. Global high-order models including gradient, Laplacian and divergence are achieved without calibration coefficients. The conservation and convergence of the discretized formulations are preserved. Therefore, the least squares MPS method is a promising approach for stability and accuracy enhancement, but more verification and validation work should be conducted to accommodate to complex flow.

2.2. Boundary conditions

2.2.1. Wall boundaries

The most widely used wall boundary condition is dummy particles that is firstly proposed by Koshizuka and Oka [3]. Several layer dummy wall particles are placed along the computation domain to supplement the number density of fluid particles. The merit of this treatment is that dummy particles are calculated in the same manner as fluid particles, and non-slip boundary is usually enforced by setting zero-velocity to these dummy particles. This approach seems to be trivial for simple geometry, such as regular flow domain, whereas it is difficult to precisely represent complex geometry such as sharp corners and curved surfaces. Past studies used two approaches to treat the arrangements of wall particles in complex flow domains. The first approach is placing wall particles in their voxel positions forming a staircase boundary. This approach can sustain particle number density at the initial value throughout the flow field, but the discretized boundary is not continuous and it may differ from the actual boundary, especially when the resolution is not high enough. The applications of this boundary treatment could be found in literatures [46–51]. The second approach is placing wall particles along the geometrical boundary of the computation domain, and dummy wall particles are placed accordingly to adhere to the geometry. This approach addresses the problem of the first approach, creating a continuous and smooth boundary, but particle number density calculation is not accurate. Li et al. [52] has compared particle number density of fluid particles along the boundary that is composed of dummy particles. The analysis indicates that the fluid particle number density is not continuous due to the shape variation of boundary, and consequently non-physical movement of fluid particles occurs because the source term of PPE is affected by particle number density. This approach has been applied by Koshizuka et al. [16], Koshizuka and Oka [53], Kouh et al. [54] and Shakibaeinia and Jin [55], in modeling a slanted plane. Moreover, applications in more complex flow boundary can be found in the studies by Nakanishi et al. [56] and Sun et al. [57,58] in modeling the geometries of Pelton bucket and cylindrical stator, respectively.

Imaginary mirror particles of the fluid particles are another popular wall boundary condition, which can be traced back to the work of Xie et al. [59]. Xie et al. implemented non-slip boundaries by setting opposite velocities of boundary fluid particles to the mirror particles. The velocities at solid surface are linearly interpolated from the velocities of the nearest fluid and mirror particles, which are always kept at the value of zero. Shakibaeinia and Jin [5] improved this method by imposing the tangential component of the fluid particle velocity on mirror particles, while the normal component is kept at constant zero. This treatment can implement slip and non-slip boundaries. Zhang et al. [26] and Park and Jeun [60] modeled the boundary with one-layer wall particles and mirror particles that are reflected from the fluid particles. Because mirror particles are corresponding to fluid particles with the normal to the wall, their positions are updated at every time step, which increases computational cost enormously. The pressures of these mirror particles are extrapolated from that of the corresponding particles, and the static pressure is also taken into account [61]. It is known that placing mirror particles in the vicinity of sharp corner is challenging because of over estimation of wall contributions. Akimoto [62] modeled mirror particles at sharp corner using a dividing line to divide inside fluid region into two parts, which could avoid the excessive and deficient generation of mirror particles. This treatment is only effective in two-dimensional simulation. In sum, the aforementioned mirror particles can maintain the physical shapes of the geometries and the slip and no-slip boundary conditions can be easily implemented. However, this treatment is only applicable to 2D and simple 3D geometries, because in complex 3D geometries the arrangement of mirror particles becomes a great challenge. Moreover, the Neumann boundary condition of pressure cannot be enforced on the wall.

To solve the problem that dummy and mirror particles cannot accurately represent complex 3D geometries, Harada et al. [63] proposed a polygon wall (PW) boundary condition. A staggered background mesh is used and the geometries are represented by these triangle meshes. The distance from a fluid particle to wall boundary is calculated by linear interpolation of the distance stored at grid nodes around this fluid particle. The Laplacian and gradient models between fluid particles and wall boundaries are modified in relation with the distance function. Zhang et al. [64] improved PW boundary condition by considering wall contributions to fluid particles using a formulation and the stabilities enhanced remarkably. Zhang et al. [65] developed a boundary particle arrangement technique and coupled it with polygon boundary condition to improve the particle number density calculation near the slopes and curved surfaces. Two sets of uniform grids are utilized in the proposed technique. The grid points in the first uniform grid construct boundary particles, and the second uniform grid stores the information of the polygon boundary condition. The wall weight function is calculated by using the newly constructed boundary particles. In [66,67] new gradient model and new PPE are proposed considering the contributions of wall part, and the pressures of wall boundaries are derived from Neumann boundary condition. Tanaka et al. [68] modeled wall geometries by using polygons together with the imaginary wall particles that are located right on the boundaries. Using this treatment the boundary conditions of viscosity and pressure can be easily enforced to the imaginary wall particles. Matsunaga et al. [69] proposed an improved wall boundary treatment for least squares MPS method. Wall geometries are expressed by surface meshes, i.e. polygons in 3D and line segments in 2D, and wall particles are dynamically generated on local boundaries. The effects of wall boundary conditions on fluid particles are considered in differential operators by Taylor series expansion. With this treatment, the slip and non-slip walls are easily applied for Neumann boundary conditions without implicit calculation scheme. Zheng et al. [70] proposed a ghost cell boundary model for wall boundary of complicated shapes. The functions of ghost cells are similar with dummy particles in the standard MPS method, but it is more flexible and can reduce computational cost.

On the other aspects, some studies tried to use wall functions instead of calculating particle number density contribution from the wall. However, these functions should be adjusted to match complex geometries [52,71]. Shibata et al. [72] developed a transparent boundary condition for the simulation of nonlinear water waves. The velocity and pressure of particles that are on the transparent boundary are set as Dirichlet boundary condition, which are calculated from analytical solution of Stokes waves. The simulation indicates that the transparent boundary condition can significantly save computation time compared with the conventional highly viscous boundary. In [73], one-layer wall particles is used for solid boundary. The contribution of dummy wall particles to particle number density in the standard MPS method is replaced by a particle number density correction function. Ng et al. [45] proposed a body-fitted boundary for complex flow domain by using unstructured pressure grids. Fluid particles are inserted to mesh centroid and boundary particle are placed at the centroid of boundary face. The formulation of PPE is modified accordingly for boundary particles.

2.2.2. Free surface

The merit of particle method is that the free surface can be easily detected using particle number density, without the presence of air particles. In the standard MPS method, the particles are defined as surface particles when particle number density around there is below βn^0. Whenever the particles are detected as surface particles, the pressure of zero is imposed on. No repulsive force is calculated among these surface particles, and thus they get very close to each other. The clustering and overlapping of surface particles can degrade the accuracy of surface calculation. Moreover, the internal particles may also be misrecognized as surface particles and the pressure of zero is enforced on. Such misrecognition will lead to pressure oscillation and calculation break down in serious conditions. Some efforts have been made to improve the accuracy of surface detection and pressure calculation.

Lee et al. [61] used both particle number density and absolute particle number N^0 for surface detection. Chen et al. [74] applied the surface particle detection technique of Lee et al. [61], and the conceptual particles are used at free surface to compensate particle number density. Tamai and Koshizuka [39] developed free surface detection algorithm in three steps, e.g. 1st step: the conventional particle number density, 2nd step: using the minimum eigenvalue of one order moment matrix, and 3rd step: scanning of neighboring particles. If the particles satisfy the following conditions, they are judged as the internal particles.

$$\begin{cases} \left\| \mathbf{x}_j - \mathbf{x}_i \right\| \geq \sqrt{2} l_0 \\ \left\| (\mathbf{x}_i + l_0 \mathbf{n}) - \mathbf{x}_j \right\| < l_0 \end{cases} \tag{19}$$

or

$$\begin{cases} \left\| \mathbf{x}_j - \mathbf{x}_i \right\| < \sqrt{2} l_0 \\ \dfrac{\mathbf{x}_j - \mathbf{x}_i}{\left\| \mathbf{x}_j - \mathbf{x}_i \right\|} \cdot \mathbf{n} > \dfrac{1}{\sqrt{2}} \end{cases} \tag{20}$$

where $\mathbf{n}$ is a normalized eigenvector associated with the minimum eigenvalue of one order moment matrix. The $\mathbf{n}$ is calculated by

$$\mathbf{n} = \frac{\mathbf{N}}{\|\mathbf{N}\|}, N = \frac{1}{n_i} \sum_{j \neq i} \left[\frac{\mathbf{r}_i - \mathbf{r}_j}{\left\| \mathbf{r}_i - \mathbf{r}_j \right\|} w(\left\| \mathbf{r}_i - \mathbf{r}_j \right\|) \right] \tag{21}$$

Duan et al. [75] applied Tamai's surface detection technique and used virtual particles to supplement the incompleteness of neighboring particles of the surface particles. The PPE and gradient model are modified accordingly by considering the contributions of virtual particles in multiphase flow. Tsuruta et al. [76] introduced space potential particles to solve inconsistency in volume conservation caused by unphysical voids. The formulations of PPE, Laplacian model and gradient model are modified with the consideration of particle–void interactions at free surface. The enhanced effect on stability is proved by simulating the tests of whirling water flow, two-phase surfacing flow, and Karman vortex.

Shibata et al. [73] proposed a surface detection method based on virtual light source and virtual screen in 2D and 3D. If the area of a particle that is illuminated by light source is larger than a threshold value, this particle is regarded as surface particle, as shown in Fig. 3. The virtual particles are assumed being on free surface to aid the derivations of PPE and gradient model, but these virtual particles are not actually located, generated or moved. With this treatment, non-zero pressure boundaries can be implemented at free surface. Sun et al. [77] used a geometrical method for surface particle detection, which is similar as Shibata's virtual light source method. The angles of connecting lines between the target particle and its neighboring particles are calculated. If there is no particle contained in the sector region that is composed by connecting lines, the target particle i is regarded as surface particle. This method is more accurate but time consuming.

As opposed to free surface detection, a moving free surface mesh to represent deformable free surface boundary was proposed by Matsunaga et al. [41]. The governing equations are separately given to the internal fluid particles and surface nodes, but their solutions are strongly coupled. The shape of free surface is explicitly defined by surface mesh, and surface nodes are dynamically added or removed to conserve fluid volume. Surface tension force acting on the surface nodes is directly calculated, without converting them to volume forces.

2.2.3. Inflow and outflow boundaries

It is not easy to implement inflow and outflow boundaries based on Lagrangian framework, because particles should flow in and out at a certain Eulerian position. Shakibaeinia and Jin [5] proposed a particle recycling strategy for inflow and outflow boundaries. An extra type of particle, e.g. storage particles, is used for particle cycling, which has no physical properties. When fluid particles flow out of the computation domain, they change to storage particles and the relevant physical properties are removed. Conversely, storage particles are changed to fluid particles and the physical properties are imposed on at inflow boundary. Fig. 4 shows the known velocity and pressure boundary conditions. Duan et al. [75] improved Shakibaeinia's inflow boundary by setting a layer of pressure particles between the ghost and fluid particles. The pressure particles involve in the pressure calculation as fluid particles. With this treatment, the pressure oscillation caused by particle type change at inflow boundary can be reduced. Given the problem of unphysical voids in channel flow, such as Karman vortices, Shibata et al. [78] optimized outflow

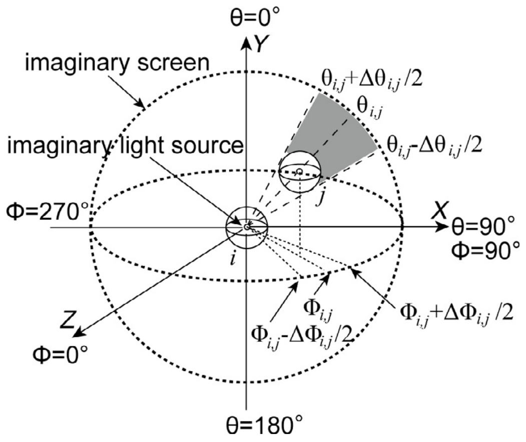
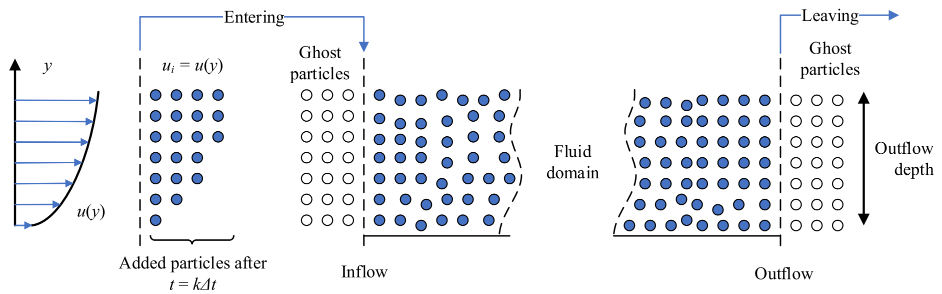


Fig. 3. Surface detection algorithm using an imaginary light source in three dimensions [73].



(a) known velocity inflow boundary condition (b) known pressure (depth) boundary condition

Fig. 4. Inflow and outflow boundary conditions [5].

boundary by imposing a fixed velocity to the fluid particles that are closing to boundary line. The outflow velocity is calculated from the velocities of upstream particles and a main criterion is to satisfy mass conservation of fluid in the channel. Nomura et al. [79] and Shibata et al. [80] adopted inflow and outflow boundaries for the analyses of melt breakup behaviors.

2.2.4. Brief summary

Several important developments have been made on free surface, solid wall, and inflow and outflow boundaries. These models are effectively applied in practical applications, and can accommodate to complicated geometries. The advanced polygon wall boundary can avoid the use of dummy wall particles, and as a result, the computation cost is reduced. Accurate detection of free surface particles improves the calculation stability. With regard to the future research points on boundary conditions, the efforts should be devoted to non-zero pressure Dirichlet boundary condition at free surface. Shibata et al. [73] has conducted primary work on this aspect, but the model is not reliable in complex flow. Secondly, the consistency, conservation and convergence of the used boundary conditions should be further studied as well. In addition, the present inflow and outflow boundaries mainly focus on velocity boundary. To satisfy the practical application the pressure inlet and outlet boundaries need to be studied in the future developments.

2.3. Surface tension

Surface tension plays an important role in the simulation of multiphase flow, such as droplet breakup and coalescence, bubble dynamics, and capillary jet breakup. Surface tension models for particle methods can generally

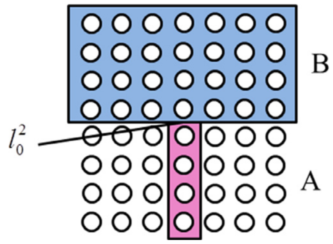


Fig. 5. Work needed to create two new surfaces with area l_0^2 [82].

be classified into two categories. The first one is based on particle–particle interaction forces, which are similar to van der Waals forces. Such models are simple and effective, but the real surface tension is difficult to be represented accurately because it depends on the intensity of particle–particle interactions. Thus, the artificial coefficients are often adjusted to reproduce the desired surface tension effect. The second one is continuum surface force (CSF) model, which is commonly used in macroscopic particle-based simulations. Surface tension is proportional to surface curvature, but the curvature is usually difficult to calculate accurately. As the local curvature is calculated as the divergence of normal vector, one can select reliable normal vectors to calculate accurate curvatures.

2.3.1. Potential based model

Shirakawa et al. [81] first developed a surface tension model based on particle internal potential force for MPS method, which is similar to molecular force. Every particle interacts with its neighboring particles and the potential force is repulsive and attractive, respectively, when the particle distance is less and larger than a certain value. The force acting on the internal particles with the same physical properties is balanced for a uniform particle distribution, while for interface particles the non-balanced potential force between different particles acts as the surface tension. However, the relation between potential force and surface tension coefficient is not given. Kondo et al. [82] built a relation between artificial constant and surface tension coefficient by the approach of creating new fluid surface, as shown in Fig. 5. The work required to detach one fluid from another fluid with a surface of l_0^2 is estimated as

$$2\sigma r_{\min}^2 = \sum_{i\in A, j\in B} P(\left|\mathbf{r}_i - \mathbf{r}_j\right|) \tag{22}$$

Kondo et al. [82] also proposed a new potential function (Eq. (23)), which is much smoother than Shirakawa's function.

$$P(r) = \begin{cases} \frac{C}{3}(r - \frac{3}{2}l_0 + \frac{1}{2}r_e)(r - r_e)^2 & (r < r_e) \\ 0 & (r \geq r_e) \end{cases} \tag{23}$$

Liu et al. [83] applied $1.5l_0$ to replace l_0 in Eq. (23) in the simulation of liquid dispersion by hydrophobic/hydrophilic mesh packing. Hattori et al. [84] simulated dynamic contact angle of a droplet sliding on an inclined surface using potential surface tension model.

2.3.2. Continuum model

In continuum surface tension model, the force is calculated on surface particles by

$$\boldsymbol{\gamma} = \sigma\kappa\delta\mathbf{n} \tag{24}$$

where σ is surface tension coefficient, κ is curvature, δ is Dirac function, and $\mathbf{n}$ is a unit normal vector at surface. Accurate calculation of curvature and normal vector is important to reproduce surface tension effect. Nomura et al. [79] proposed a method to calculate surface tension in 2D model. The curvature and normal vector are obtained by computing the distribution of particle number density, which is equivalent to fitting local interface by an arc of a circle.

$$\kappa = \frac{2\cos\theta}{r_e^{st}} \tag{25}$$

$$2\theta = \frac{n_i^{st2}}{n_0^{st}}\pi \tag{26}$$

$$\mathbf{a}_i = \frac{n_i^{+x} - n_i^{-x}}{2l_0}\mathbf{n}_x + \frac{n_i^{+y} - n_i^{-y}}{2l_0}\mathbf{n}_y \tag{27}$$

$$\mathbf{n}_i = \frac{\mathbf{a}_i}{|\mathbf{a}_i|} \tag{28}$$

where n_i^{st2} and n_0^{st} are particle number densities calculated using different weight functions, which are equal at a flat surface ($2\theta = \pi$); n_i^{+x} and n_i^{+y} are particle number densities at four positions near particle i. Because of the simplicity of Normura's surface tension model, it is adopted in the simulations of drop impacting onto liquid film [85], single bubble rising [86], and droplet generation in micro flow [87]. Ichikawa and Labrosse [88] extended Nomura's surface tension model to 3D by calculating the normal vector with the assistance of SPH scheme. The calculated normal vector is less noisy than that obtained with Normura's scheme. Park and Jeun [89] tried to calculate surface normal vector from the divergence of particle number density, but no remarkable difference is observed in the verification of liquid drop oscillation.

Other than calculating surface curvature and normal vector using particle number density, an alternative approach is to calculate normal vector using color function. The value of color function c is defined as 0 in one phase and it is 1 in other phase. The normal vector is calculated from the gradient of color function at interface.

$$\mathbf{n} = \frac{\nabla C}{|\nabla C|} \tag{29}$$

The curvature can be written as a divergence of normal vector as

$$\kappa = \nabla \cdot \mathbf{n} \tag{30}$$

This color function approach is much suitable for multiphase flow, because the gradient of color function should be evaluated between two phases. When calculating surface tension at free surface, air particles are necessary to be arranged. Duan et al. [90] developed a contoured continuum surface force model for 2D and 3D simulations, in which the local contour is calculated from smoothed color function. The curvature of local contour is analytically calculated on the basis of Taylor series expansions. Smooth radius has a significant effect on the accuracy of surface tension calculation. If the smooth radius used is small, the calculated curvature fluctuates heavily; conversely, if a large smooth radius is used, the sharp corners on the interface may be smoothed away. Thus, an appropriate smooth radius should be calibrated before application. Fig. 6 shows the calculated color function and contours with different smooth radii. Sun et al. [77] reconstructed free surface based on level set function, and surface tension force is calculated by using the equation as follows.

$$\mathbf{F}_{s,i} = \sum_j \frac{\sigma\kappa_j \mathbf{n}_j W_{ij}}{\eta_j} \cdot \frac{\rho_i \Delta x_i}{m_i} \tag{31}$$

where the weight coefficient η_{j} is calculated by $\eta_j = \sum_k W_{jk}$.

2.3.3. Brief summary

The existing surface tension models are potential based model and continuum model. The potential based model can be easily implemented in simulations, but the tension calculated at two fluid interface is a relative force, and thus surface tension coefficient should be calibrated in every practical calculation. The prevailing continuum surface tension model is based on color function of different particle types. The precise calculation of interface curvature is a great challenge. Different curvature calculation models such as by using particle number density, calculating divergence of normal vector, and using contours of color function are successively developed, but the curvatures are usually filtered at local corners. Therefore, the studies should be continued on the precise calculation of interface curvatures with special attention being paid on sharp corners. Surface tension model should be applied together with stability and accuracy improvement techniques to ensure a continuous and consistent interface.

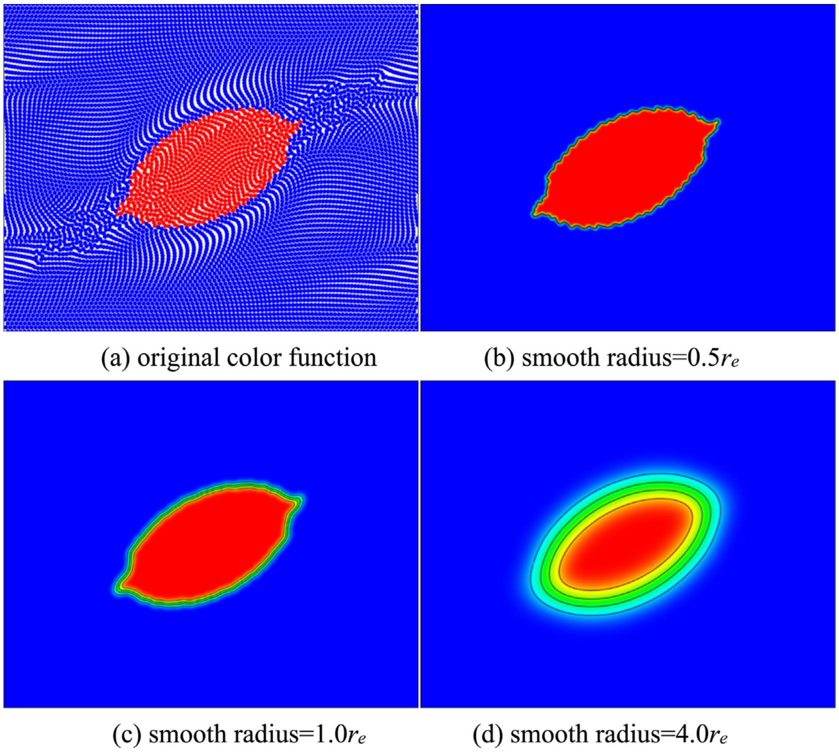


Fig. 6. smoothed color function and contours by different smooth radii ($r_e = 3.1 l_0$) [90].

2.4. *Multiphase flow*

2.4.1. *Gas–liquid two-phase flow*

Multiphase flows are ubiquitous in nuclear engineering, ocean engineering and other industrial processes, which have the characteristics of unsteady state, multi-physics and presence of moving and large deforming boundaries. The most challenging issues in numerical simulations of multiphase flows are free surface and interface capturing and physics discontinuity at phase interface, including density and viscosity. Particle method, using particles to represent fluid, has flexibility and robustness in dealing with largely deformed moving boundaries. It is not necessary to introduce special treatment for interface tracking and reconstruction. As each particle possesses its own physical properties, an inevitable problem that must be encountered is the discontinuities of fluid density and viscosity at interface. The discontinuities in physics may cause mathematical discontinuity and accordingly a discontinuous acceleration field at phase interfaces. These discontinuities may lead to instabilities and even end up of calculations. Researchers have made some efforts to solve the discontinuity problems that arise from density and viscosity differences between two different particles. These approaches include density smoothing schemes, density averaging, background pressure, artificial repulsive pressure force, and artificial surface tension force.

Koshizuka et al. [91] and Ikeda et al. [21] calculated gas and liquid pressures in two steps. In the first step, the pressure of liquid particles is calculated by regarding the interface with vapor as free surface, and then the pressure of gas particles is solved in the second step by regarding liquid particles as rigid wall. There is no density smooth treatment at gas–liquid interface. Similarly, without using density smooth technique, Shimizu et al. [92] proposed an approach to deal with large density drop at air–water two-phase flow based on the concept of space potential particles. The pressure calculation is also divided into two steps as illustrated in Fig. 7. The pressure of air particles is calculated by considering the velocity of water particles as velocity boundary; then, the calculated pressure of air particles is mirrored to the space potential particles of water particles in the vicinity of interface and serves as pressure boundary in the calculation of water phase. The proposed method is coupled with some stability and accuracy improvement techniques in the verification simulations of sloshing flow and oscillating drop under

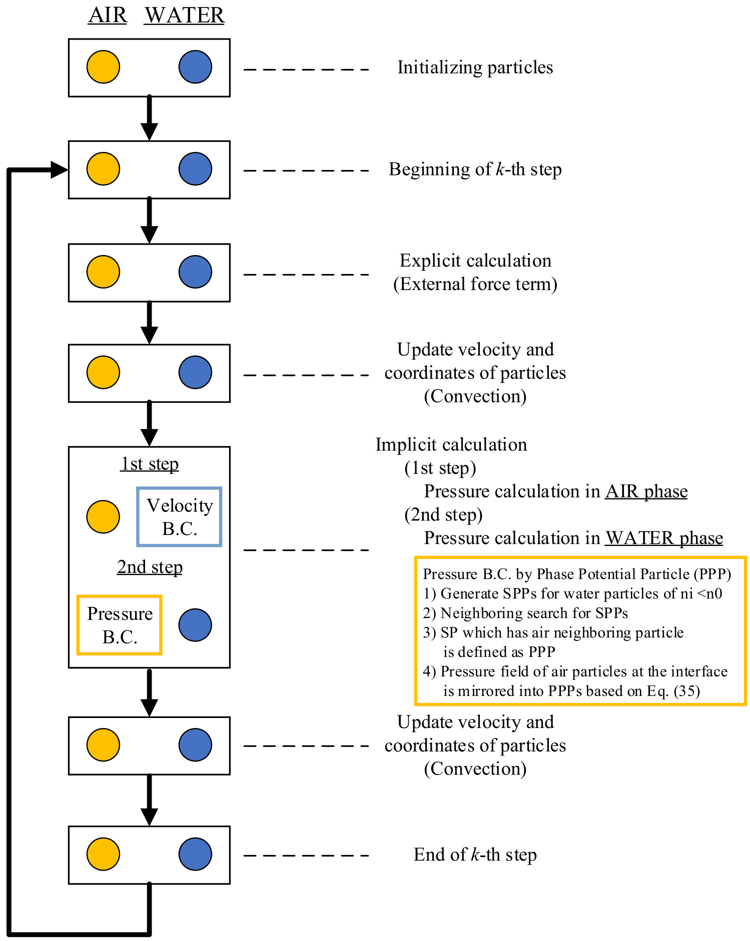


Fig. 7. Algorithm of multiphase flow simulation [92].

a central force field. Khayyer et al. [37] applied OPS scheme (mentioned in Section 2.1.4) to the improved MPS method (e.g. high-order differential operators with dynamic stabilization) to simulate multiphase flow with large density ratio. The methods mentioned above do not use any density smoothing technique at phase interface. Using these methods, the material discontinuity and pressure continuity at interface are preserved.

Shakibaeinia and Jin [93] developed a method for multiphase flow simulation by treating the system as multi-density multi-viscosity fluid, in which the weakly-compressible formulations are used to solve a single set of equations for all phases. The density used in pressure calculation equation is smoothed and different viscosity average schemes are evaluated. The pair-wise particle collision model is also modified on the basis of Koshizuka's collision model by considering the multiphase force that arises due to density and viscosity differences. Natsui et al. [94] applied similar density smoothing model for particles in the transition region and used explicit pressure calculation algorithm to the simulation of air bubble rising in water. Khayyer and Gotoh [95] improved the commonly applied density smoothing scheme by using a first-order accurate Taylor series expansion of density (Eq. (32)) for modeling an accurate and consistent phase interface.

$$\rho_i = \frac{1}{\sum_{j\in I} W_{ij}} \sum_{j\in I} \left(\rho_j - \frac{\partial \rho_i}{\partial x_{ij}} x_{ij} - \frac{\partial \rho_i}{\partial y_{ij}} y_{ij} \right) W_{ij} \tag{32}$$

The newly proposed density smoothing scheme considers the spatial contributions of densities at neighboring particles and the spatial variations of density at target particle itself. A Taylor series consistent pressure gradient model is also developed to compensate the error caused by non-regular particle distribution. The method is proved to be effective in preserving the sharpness of spatial density variations and provides a continuous pressure field by minimizing unphysical perturbations. Khayyer et al. [37] evaluated the schemes of OPS and density smoothing for multiphase flows with large density ratio, and concluded that the improved MPS with OPS scheme outperform density smoothing in terms of accuracy, convergence and conservation properties. Duan et al. [24] analyzed the force and acceleration in the interactions of light and heavy particles and pointed out that the large acceleration of light particles is the cause of instability. The authors proposed two approaches to stabilize the simulations with large density ratio: multiphase MPS with harmonic density (MMPS-HD) and multiphase MPS with continuous acceleration (MMPS-CA). In MMPS-HD method, harmonic density is applied in multiphase PPE to reduce the exceptionally high acceleration at light particles (Eq. (33)). In MMPS-CA method, new stable multiphase formulations are derived based on the locally weighted average of particle accelerations to preserve the continuity of acceleration and velocity at interface. The particle stabilizing term is decoupled from original pressure gradient model, as shown in Eq. (34).

$$\frac{2d}{n^0\lambda}\sum_{j\neq i}\frac{\rho_i+\rho_j}{2\rho_i\rho_j}(P_j^{k+1}-P_i^{k+1})w(\left|\mathbf{r}_j-\mathbf{r}_i\right|)=\frac{1-\gamma}{\Delta t}\nabla\cdot\mathbf{u}^*-\frac{\gamma}{\Delta t^2}\left(\frac{n^*-n^0}{n^0}\right)+\alpha\frac{1}{\Delta t^2}P_i^{k+1} \tag{33}$$

$$\left\langle\frac{1}{\rho}\nabla P\right\rangle_i=\frac{d}{n^0}\sum_{j\neq i}\left\{\frac{2(P_j-P_i)(\mathbf{r}_j-\mathbf{r}_i)}{(\rho_i+\rho_j)\left|\mathbf{r}_j-\mathbf{r}_i\right|^2}\mathbf{C}_{ij}w_{ij}\right\}+\frac{d}{n^0}\sum_{j\neq i}\left\{\frac{(P_i-P'_{i,\min})(\mathbf{r}_j-\mathbf{r}_i)}{\rho_i\left|\mathbf{r}_j-\mathbf{r}_i\right|^2}\mathbf{C}_{ij}w_{ij}\right\} \tag{34}$$

where the minimum pressure $P'_{i,\min}$in Eq. (34) is only searched among the neighboring particles that have the same phase with particle i. In addition, Duan et al. [15] also tried to use the corrective matrixes for all the particle interaction models and applied the modified formulations to the simulation of underwater oil spill from the damaged tanker.

Liu et al. [96] proposed a hybrid particle–mesh method for viscous and incompressible multiphase flows. The sharp interface with density and viscosity jump is preserved by representing one phase with moving particles and the other phase with stationary mesh, as illustrated in Fig. 8. The density, viscosity and surface force calculated on interfacial particles are extrapolated to mesh (Fig. 9), and then the velocity and pressure fields are calculated on mesh by finite volume method. The motion of particles is realized by interpolating the velocity from mesh. The continuum surface force model is applied to evaluate phase interaction force and wall adhesion force. This method does not use any smoothing treatment for interface interaction, and enables more accurate and faster convergent modeling of 2D and 3D multiphase flows. Erkan et al. [97] developed a VOF-MPS method by combining VOF solver of OpenFOAM code with MPS method. The surface tension calculation of VOF model is introduced to MPS method. The gas–liquid interface is defined by particle distribution and the physical parameters of liquid volume and velocity vector are transferred between grids and particles. 3D VOF-MPS method provides a smoother liquid–gas interface in the simulation of droplet deposition into a thin liquid film. Matsunaga et al. [98] developed a grid–particle hybrid method for fluid mixing simulation. The hybrid method calculates the fluid flow using grid method and solves the advection and diffusion of chemical species using MPS particle method. To improve the smoothness of fluid interface and deal with large density difference at interface without tuning of calibration parameters, Ng et al. [99] developed a moving particle level-set method, which enforces the divergence-free condition to background mesh and treats particles merely as observation points without mass property. A new particle density interpolation scheme is proposed by using the conservative level-set method and mass conservation at interface is achieved.

Moreover, some artificial inter-force between particles with large density and viscosity difference is applied in multiphase flow as well. Wang and Zhang [100] applied the Peridynamic theory to MPS method to study multiphase flow with discontinuity. The stress tensor at fluid interface is solved by MPS method. The integral form of governing equation can reduce the influence of discontinuities of fluid density and viscosity on the accuracy and stability. In addition, the divergence decomposition of stress tensor and harmonic mean density are adopted to reduce the influence of density discontinuity on interface calculation. A modified matrix is also added to decomposition equation to deal with the situation with irregular particle distribution. With the improved method, a smoother interface is obtained and the problem of particle penetration at interface is effectively avoided. Kim and Kim [101] extended MPS method to a multiphase system with multiple interfaces by including self-buoyancy-correction, surface tension, and interface boundary condition models. The modified MPS method is used to the simulation of Kelvin–Helmholtz instability arose from Poiseuille's flow.

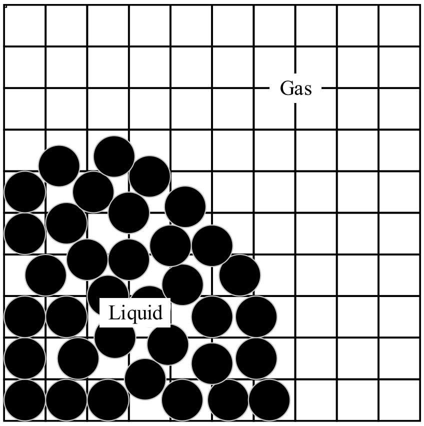


Fig. 8. The setup of mesh and particle [96].

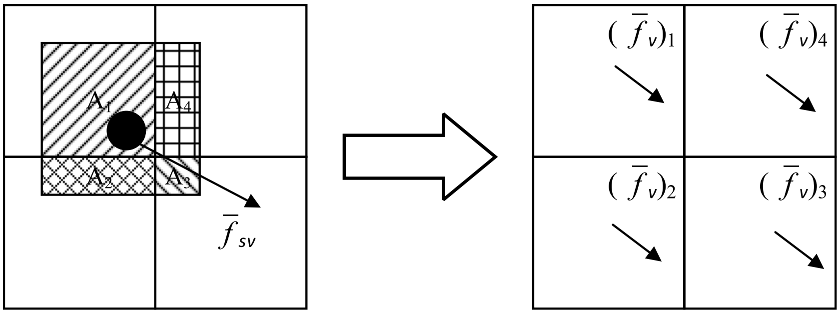


Fig. 9. Extrapolation of surface force from particle to mesh [96].

2.4.2. Solid–liquid two-phase flow

The approaches of numerical simulation of solid–liquid flow can be categorized as continuum and discrete descriptions of the solid granular material. The continuum description of granular flow treats the solid body as a complex fluid (e.g. non-Newtonian fluid) and solves the conservation equations of mass and momentum to predict the state of flow system. The contact forces of solid debris are not considered. In contrary to the continuum description of granular flow, the discrete description calculates macroscopic and microscopic forces acting on the individual granular particles. In this category, the discrete element method (DEM) [102] is widely used in the simulation of granular flows, which evaluates the contact forces based on Newton's law. In combination with MPS method, the solid–liquid flow can be effectively predicted, where the DEM calculates the motions of submerged granular flow and the MPS method calculates continuum liquid flow.

Koshizuka et al. [16] first proposed a passively moving model for calculating solid body motion in fluid. The solid body is represented by particles with fixed relative configuration. The solid particles are calculated using the same procedure as fluid particles at first, and then additional procedure is applied to solid particles to calculate the transition and rotation velocity vectors of solid body. This passively moving model only considers the pressure force from ambient fluid. Nodoushan et al. [103] modeled multiphase system of granular material and ambient fluid by treating it as a multi-density and multi-viscosity system, and a single set of governing equations is solved for the entire flow field. The effective viscosity of granular phase is predicted using a regularized Herschel–Bulkley rheological model with a pressure-dependent yield criterion. The weakly compressible scheme is applied for the stability of calculation. Many adjusting coefficients are adopted in the calculation models of effective viscosity and pressure, and they are calibrated in the simulations of dry and submerged granular flows. Tajnesaie et al. [104] replaced the Herschel–Bulkley rheological model in Nodoushan's modeling by a new visco-plastic rheological model, which is a function of dynamic inter-grain mechanical pressure. The results present more accurate granular fields. Nabian and Farhadi [105] proposed a multiphase MPS method by treating the multiphase system as a multi-density and multi-viscosity fluid, and it solves one set of hydraulic equations for solid and liquid materials. The granular media are assumed to behave as non-Newtonian fluid and the density and viscosity are averaged at phase interface. Kim [106] simulated mobile-bed behavior induced by dam break using the modified MPS method, where the self-buoyancy and surface tension models are considered in multiphase particle interaction models.

Park and Jeun [60] coupled rigid body dynamics with MPS method to simulate solid–liquid phase interaction forces. MPS calculation is performed for all the particles that are considered to have no mass or volume at first, and then the rigid body dynamics of solid particles are calculated by considering linear momentum and angular momentum conservations. Sakai et al. [71] and Yamada and Sakai [107] developed a MPS-DEM method to simulate solid–liquid flows involving free surface, in which the motion of liquid phase is modeled with MPS method and the solid phase is solved using DEM by considering the drag force, contact force, gravitational force, virtual mass force, lubrication force, and pressure force. The solid and liquid phases interact with each other through these forces. The models are validated by simulating the solid–liquid flow in a rotating cylindrical tank in two and three dimensions. The influences of hydrodynamic forces on the macroscopic behavior of solid particles are investigated. Harada et al. [108] introduced large eddy simulation turbulence model to solid–liquid two-phase flow. The DEM is applied to calculate solid particle forces in the framework of Euler–Lagrange. The sedimentation process of a particle cloud in water is simulated, and the wake induced by settling particles is presented by using smaller grid scale than particle diameter. In Harada et al. [109], the drag force acting between solid and fluid phases is described with voidage function, and the momentum balance in the interaction of solid and fluid phases is consistently conserved. Wang et al. [110] incorporated passively solid moving model and DEM to explicit MPS method to simulate solid–fluid and solid–solid interactions, respectively. The different time steps are used in computation algorithm.

2.4.3. Brief summary

In the simulations of gas–liquid two-phase flow, the challenges are large density difference at interface and the instability arising from interface. The forces acting on the light and heavy particles are equal, but the accelerations are considerably different due to density difference. The extremely large acceleration of light particles is the cause of instability. Through the review of past studies, the treatments of discontinuities at gas–liquid interface can be categorized into three types: calculating two phases separately, density smoothing, and particle–grid hybrid method. The approach of separately calculating two phases can preserve the sharpness of interface; that is, the results are more closed to the physical conditions. However, this treatment is only effective for simple two-phase flow phenomena, and violent instability may present in complex conditions. Density smoothing is a generic approach, which is also applied in grid method. The specific smooth treatments include local average and Taylor series expansion considering the contribution of neighboring particles. They can be used in various complex conditions without considering the geometry of interface, but the expense is losing the sharpness of interface. particle–grid hybrid method is another approach, in which the gas and fluid are represented by grid and particles, respectively, and the physical parameters are interpolated and extrapolated between grid and particles. Continuous pressure and discontinuous density at interface are realized, but this method is very time consuming.

Regarding to the solid–liquid two-phase flow, continuum and discrete descriptions of solid granular material have been investigated. The continuum approach conceives the solid–liquid flow as a multi-density and multi-viscosity system, and a single set of governing equations is solved for the entire flow field. This approach cannot characterize the actual motions of solid particulates, because the interaction forces, especially the contact force of solid particulates, are not accurately calculated. The discrete approach, represented by MPS-DEM coupling method, is a promising method for solid–liquid flow. MPS-DEM method solves fluid flow using MPS method and solves hydraulic forces and various forces in solid particulates using DEM. The linear and angular momentum conservations are preserved for every particulate. An attention is that different time steps should be set for MPS and DEM algorithms, because DEM usually requires a very small time step.

2.5. Fluid–structure interactions

Fluid–structure interactions (FSI), including the deformation of elastic structures, can be encountered in nuclear engineering, ocean engineering, and other industry processes. As the development of computation technology, a number of numerical studies have devoted to the simulations of FSI problems. Most of numerical analyses for FSI problems are conducted in Eulerian framework with grids. Using grid-based method, it needs special boundary treatment for large deformation problem and additional solver algorithm to regenerate grid configuration. However, these special treatments may degrade mass and momentum conservations at fluid and solid interface. The particle-based Lagrangian method can easily deal with fluid large deformation problems. Recently, SPH and MPS methods have been applied to fluid–structure interaction problems by using fully Lagrangian particle method or combining them with grid method.

Lee et al. [111] proved the feasibility of fluid-shell structure interaction analysis by using a particle and finite element coupled method, where MPS method is used to analyze fluid flow and MITC4 shell element is used to FEM analysis of structure. The fluid solver uses a semi-implicit algorithm and the structure solver uses an explicit algorithm. A partitioned coupling scheme is used between the fluid and structural solvers. However, numerical results in the simulation of sloshing in elastic walled tanks is not stable, where fluid particles present unphysical splash. Hwang et al. [112] developed a fully Lagrangian method for the simulation of fluid–structure interaction based on MPS particle method and elastic structure models. In the calculation of fluid flow, the external force from structure particles is considered in momentum conservation equations, and the calculated position and velocity of structure particles at previous time step serve as boundary conditions for the calculation of fluid particles. The hydraulic force acting on structure is incorporated in the governing equations of elastic solid. The coupled modeling is applied to elastic plates that subject to time-dependent hydrodynamic pressures corresponding to a dam break and a violent sloshing flow. Fig. 10 shows the fluid–structure interaction in an elastic plate subjecting to dam break. In [113], the authors improved the particle-based fluid–structure interaction solver by considering structural particles at the same elevation as one group. As a result, the force acting on a particle belonging to one group is calculated by dividing total force by the total mass of group. A smooth force distribution on elastic baffle is gained under the action of fluid, as presented in Fig. 11. Khayyer et al. [114] further improved the consistency at fluid–structure interface through the consideration of mathematical basis of MPS method, and the multi-resolution scheme is applied to reduce computation cost. Falahaty et al. [115] improved particle method by using stress point integration for the simulation of fluid-nonlinear elastic structure interaction. Nonlinear structure models using nodal integration (referred as moving least squares model) and stress pint integration (referred as dual particle dynamics model) are coupled with MPS method, respectively. The coupled methods are comparatively validated in the benchmark tests of hydrostatic water column on elastic plate, dam break with an elastic gate, hydro-elastic slamming of marine panels, and violent sloshing flows in tanks. The comparisons indicate that the combination of MPS method with dual particle dynamics model, together with the modified interface boundary condition, shows remarkable improvements in suppressing spurious oscillations related to zero-energy modes.

Sun et al. [116] enhanced the stability of FSI calculation by using error compensating part to PPE source term, new type of solid and free surface boundaries, and the PS technique. Sun et al. [117] further developed a coupled model of rigid body motion and modal superposition to study the interaction between violent water and flexible structure, in which the modal superposition model calculates the relatively small elastic deformations of the flexible structure. The mutual effect between rigid body motion and elastic deformation is considered, which affects the rigid body motion pattern. The fluid motion is solved with the modified MPS method that effectively suppresses pressure fluctuation. The proposed method is verified in the simulations of two nonlinear problems, i.e. breaking-water-dam impacting a floating beam and flexible wedge slamming into water, with good agreements with the experimental results. Zhang et al. [118] investigated the interaction between waves and free rolling body with the modified MPS method including the wave-maker module and free roll motion module. Sun et al. [119] proposed a FSI solver by coupling MPS method and DEM, and the DEM is improved by introducing a bonded-particle model to predict structure deformation. A deformation coefficient is introduced to evaluate the changes of structure deformation and displacement. In addition, the refined interface force calculation and multiple time-step algorithm are applied in the combination of MPS and DEM.

In summary, fluid–structure interactions are modeled in the above mentioned studies, but the work on this aspect is preliminary. The test cases for model verification are simple. Even so, the developed models provide a solid foundation for future studies on the stability, accuracy and consistency of fluid–structure interaction simulations. Fluid–structure interaction models are expected to be applied in complex conditions, such as structure fracture mechanics, dynamics of composite structures, and violent fluid flow.

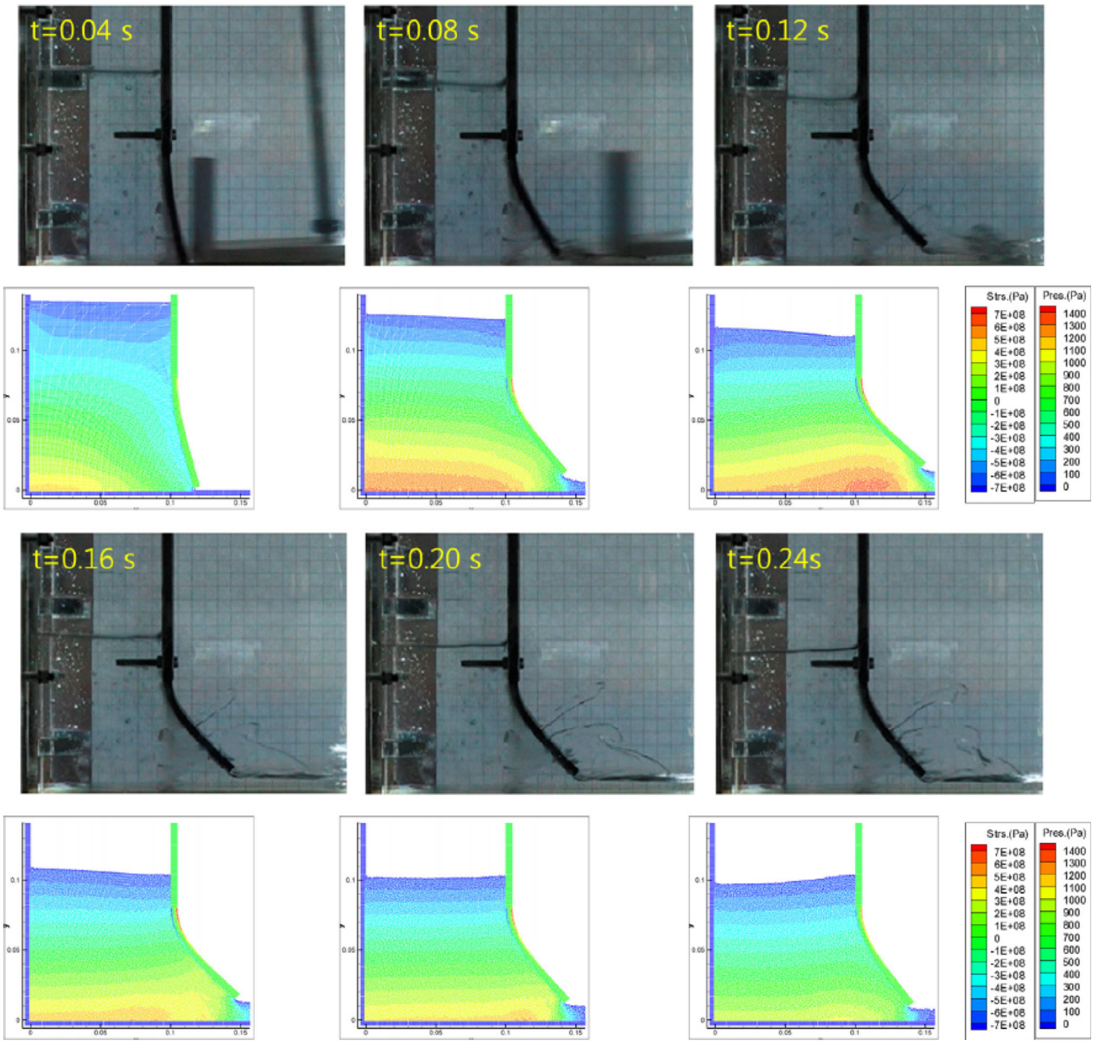
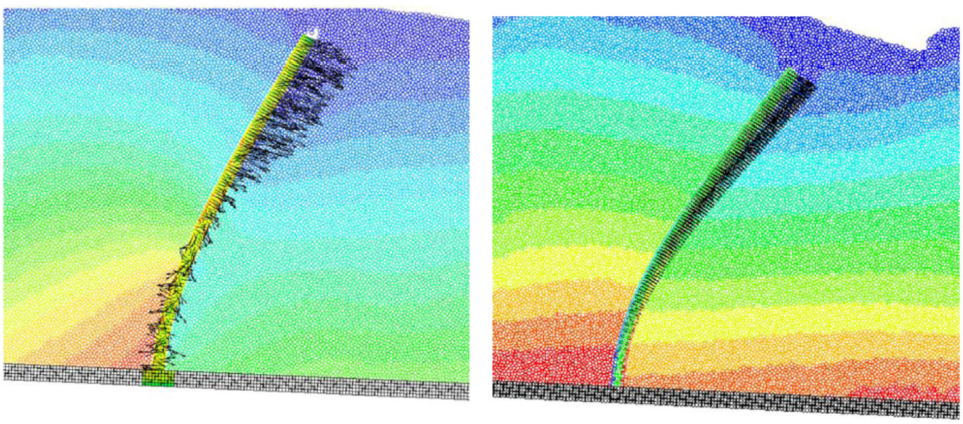


Fig. 10. Fluid–structure interaction in an elastic plate subjecting to dam break [112].



a) previous method b) modified method

Fig. 11. The force distribution on the elastic structure calculated by (a) the previous method [112], and (b) the improved method [113].

2.6. Multi-resolution technique

Particle method is different to mesh method that can use different size grids at different regions. Several attempts have been tried to use multi-resolution scheme in SPH method. However, multi-resolution scheme for MPS method is more difficult to implement because the PPE is solved implicitly. Traditional techniques of keeping particle number density constant and detecting surface particles based on the deficiency of particle number density cannot be applied anymore. Recent years, some noteworthy developments are reported for MPS method to use different spatial resolutions.

Shibata et al. [120] developed an ellipsoidal particle model to reduce computational cost. Particle distances at x and y directions are different. A coordinate transformation scheme is proposed to treat ellipsoidal particles in the same manner as sphere particles. Laplacian model in the original MPS method is modified to make it applicable to ellipsoidal particles. Chen et al. [121] developed a multistep particle splitting and coalescing scheme. One particle splits into seven daughter particles by five steps, which enables the daughter particles to find their proper positions. However, the merging only takes places between two small particles. The volume and momentum are conserved during particle splitting and merging processes. Moreover, a cubic spline kernel function is used to calculate the contributions from the particles with different sizes, and a new gradient model is developed to minimize first-order error associated with different particle sizes. Shibata et al. [122] developed an overlapping particle technique for multi-resolution simulation of particle method. The computation domain is divided into several sub-domains and the adjacent sub-domains partially overlapped with each other. Different particle sizes are employed for each domain and the pressure and velocity information is interpolated at the overlapping regions of sub-domains, as illustrated in Fig. 12. The particles only interact with other particles in the same sub-domain, and the calculation results at one time step are assigned to other sub-domain as a boundary condition of inlet/outlet boundary. The overlapping particle technique can avoid the error and instability caused by the interactions of different size particles because particle size is uniform in one sub-domain. Tang et al. [123] applied multi-resolution technique to the simulations of violent free-surface flows. An averaged effective radius is used to the interaction of two particles with different sizes. In the simulations, however, particle splitting and merging are not involved. Tanaka et al. [68] developed multi-resolution technique based on the least square MPS method. A parameter of packing ratio is used to replace particle number density and new formulations for differential operators are derived. To keep the accuracy and stability, one particle is allowed to split into a maximum of two particles with the conservation of particle volume, and vice versa. Novel wall and pressure boundary conditions are developed as well. Multi-resolution scheme is verified by the simulations of Poiseuille flow, multi-diameter channel flow, Taylor green vortex, and the flow in gears. Khayyer et al. [114] applied multi-resolution scheme to incompressible fluid-elastic structure interactions in ocean engineering. The formulations of effective radius, weight function and particle number density are modified and the concept of potential number density is used to enhance volume conservation at fluid–structure interface, to ensure consistency of particle-based discretization, and to enhance the imposition of boundary conditions. Some other advanced schemes, such as error-compensating source, higher-order Laplacian, and gradient corrective matrix, are also applied to improve the stability and accuracy of the simulations. Moreover, Yoon et al. [124,125] applied multiple particle resolution to a particle–grid hybrid method. In the simulation of bubble generation, fine resolution is used at gas–liquid interface and this fine resolution follows bubble motion, as show in Fig. 13.

In a summary of above studies, multi-resolution technique has not got sufficient development in MPS method, which is an effective approach to reduce computation cost. The difficulties are particle number density calculation that varies according to the scales of neighboring particles and the particle volume that cannot be considered as constant. Particle interaction models should be revisited by considering volume difference of large and small particles. In the algorithms of splitting large particles and merging small particles, the focus is the jump of particle number density and thus the effective radius of particle interaction model should be averaged or smoothed. Instability is still a challenge in multi-resolution scheme, and the consistency, conservation and convergence in the interaction of particles with different sizes are important research topics.

3. Application in nuclear engineering

3.1. Fundamental thermal hydraulics

Yoon et al. [125] simulated bubble growth, departure and deformation processes in nucleate pool boiling using a mesh-free numerical method (MPS-MAFL). The liquid phase is modeled using moving particles but the bubble is not

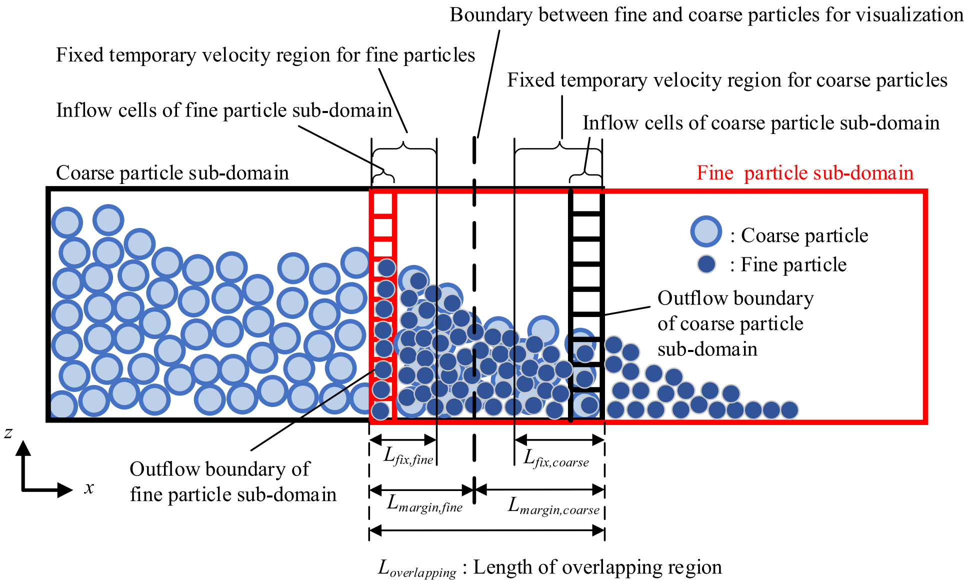


Fig. 12. Conceptual image of overlapping particle technique [122].

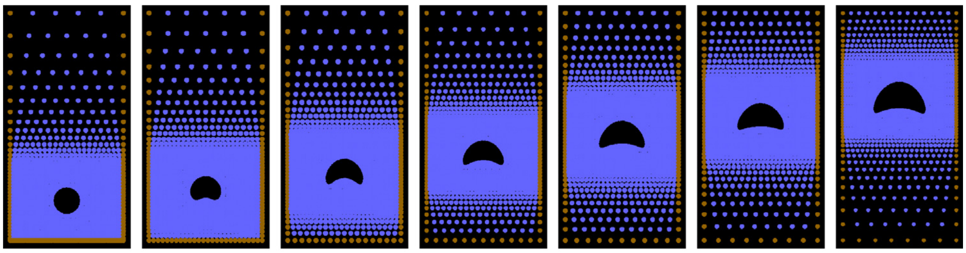


Fig. 13. Numerical simulation of rising bubble [125].

represented with particles; the interface of liquid and bubble is tracked by the movable positions of interfacial liquid particles. The results indicate that bubble departure diameter is in proportion to contact angle and the square root of surface tension. The simulated bubble growth rate, departure radius and heat transfer rate show good agreements with the experimental results. Heo et al. [126] investigated bubble growth in transient pool boiling with high heat flux and high subcooling conditions using the MSP-MAFL method. It is found that initial bubble radius has little effect on growth initiation time and bubble departure radius. Chen et al. [127] investigated the effect of bulk liquid velocity, liquid subcooling, wall superheat and surface incline angle on bubble dynamics including bubble shape, liftoff diameter, and liftoff time in flow boiling conditions. Tian et al. [128] applied MPS-MAFL method to the simulation of bubble condensation in water. Bubble deformation, lifetime and bubble size history under lower and higher degrees of liquid subcooling and different system pressures are analyzed. Li et al. [129] and Zuo et al. [130] simulated argon and nitrogen bubble rising behaviors in liquid metal, which expand MPS method to the thermal hydraulics research of advanced reactor. Chen et al. [131] reviewed the achievements on bubble dynamics analysis using MPS method.

Xie et al. [132] investigated liquid drop deposition in annular-mist flow regime of boiling water reactor (BWR) to analyze the possibility and effect of splash occurrence. Shirakawa et al. [81,133,134] studied the flows around BWR spacer, subcooled boiling with a special interest in the location of void departure, and the void distribution in a circular tube by using a two-fluid particle interaction model in the framework of MPS method. Ui et al. [135] developed a solid–liquid multiphase model to simulate debris transport, settling and resuspension in the case of loss-of-coolant accident of a pressurized water reactor (PWR). The model can solve the motion of solid particles with different sizes. Turbulence model is also coupled with MPS method. With the developed method, the analysis of debris transport process in a full-scale PWR containment vessel floor is carried out. Xiong et al. [136] simulated droplet impingement onto a rigid wall in two and three dimensions, which could lead to pipe wall thinning. The impact pressure and shear stress distributions are sensitively evaluated under different droplet diameters and impact obliqueness. Xiong et al. [137] further investigated the effect of water film on droplet impingement damage. The comparison with dry wall condition indicates that water film can effectively mitigate or even eliminate high peak pressure that appears at the contact edge. Wang et al. [110,138] simulated water-flooding processes in nuclear reactor building of AP1000 with explicit MPS method, in which the floating bodies are considered.

3.2. Melt behavior in severe accident

Ikeda et al. [139] developed boiling and solidification models for particle method to analyze molten core behavior in nuclear reactor severe accidents. For solidification model in MPS method, when the temperature of a liquid particle decreases to freezing point, it is defined as a mixture particle and the physical properties of mixture particles are calculated based on solid fraction. With the developed boiling and solidification models, the injection of molten corium into a water pool is calculated under various conditions and the results are compared with FARO-L14 experiment. Xiong et al. [140] incorporated an enthalpy-based viscosity model to phase transition process to realize a smooth viscosity transition at liquid–solid interface, and an implicit computation algorithm of viscous term is applied. Ikeda et al. [21] conducted a preliminary analysis on the dynamics of water jet injecting into a pool of denser liquid under non-boiling and isothermal conditions, to investigate coolant injection into a melt pool. Shibata et al. [141] investigated jet breakup behavior by ignoring surrounding continuous fluid. The effects of Weber and Froude numbers on jet breakup length and droplet size distribution are investigated, and the results agree well with the experimental data and correlations. Nomura et al. [79] investigated melt drop breakup behavior in coolant pool during steam explosion. The boiling of water and the solidification of melt drop are not considered, but the effect of steam film surrounding melt drop is studied. It is found that steam film can suppress the breakup of melt drop. In addition, the critical breakup Weber number of a melt drop in the condition with steam film is approximately 50. Duan et al. [142] investigated the deformation and breakup processes of a UO_2 drop in coolant pool at around critical Weber number with the neglect of steam film. Melt drop breakup process can be divided into two stages: the stage of drop stretching and thinning normal to flow direction and the stage of detach points appearing on the surface of drops due to the growth of surface waves. The detached points presented on the stretched melt drop increase with the increase of Weber number. Li et al. [143,144] investigated size distribution and surface area of the fragmented melt during breakup process by considering melt drop with and without steam film, as shown in Figs. 14 and 15. The results indicate that steam film has insignificant effect on fragment amount and contact boundary. Park et al. [145] simulated debris mixing and sedimentation processes after the fragmented melt solidifies, which are related to debris bed formation and coolability. MPS method is coupled with a rigid body dynamic model to simulate the motion of debris. The developed method allows to identify key characteristics, such as debris jet penetration, water leveling, and final debris configuration, in debris sedimentation and debris bed formation. The comparisons of 2D and 3D simulations indicate that 2D simulation is insufficient to predict debris sedimentation process, because debris distribution depends on particle agglomeration and mixing prior to debris settling. Different with Park's debris modeling method, Muramoto et al. [146] modeled debris using the clusters of particles and evaluated fuel debris criticality in water by coupling MPS method with MVP code.

Kawahara and Oka [147] simulated molten core spreading behavior on ex-vessel floor and compared melt front distance with FARO L26-S experiment. Yasumura et al. [148] investigated the influence of crust formation on corium spreading behavior by analyzing VULCANO VE-U7 experiment. Duan et al. [149] proposed a new viscous term solution algorithm for MPS method to simulate melt crust formation by raising viscosity, as shown in Fig. 16. The proposed solution algorithm can eliminate numerical creeping when solving high viscous flow. In the further analysis of VULCANO VE-U7 spreading in [150], the effect of contact resistance at melt-substrate interface on heat transfer and the influence of agitation effect of gas bubbles on melt spreading flow are modeled. Different melt spreading distances on ceramic and concrete substrates are predicted and the mechanism of such difference is ascribed to the effect of gas bubbles generated during concrete decomposition.

Li and Oka [151] analyzed molten corium-concrete interaction (MCCI) with the phase transition modeling of melt and concrete. The developed model is applied to the simulations of SURC-2 and SURC-4 experiments and

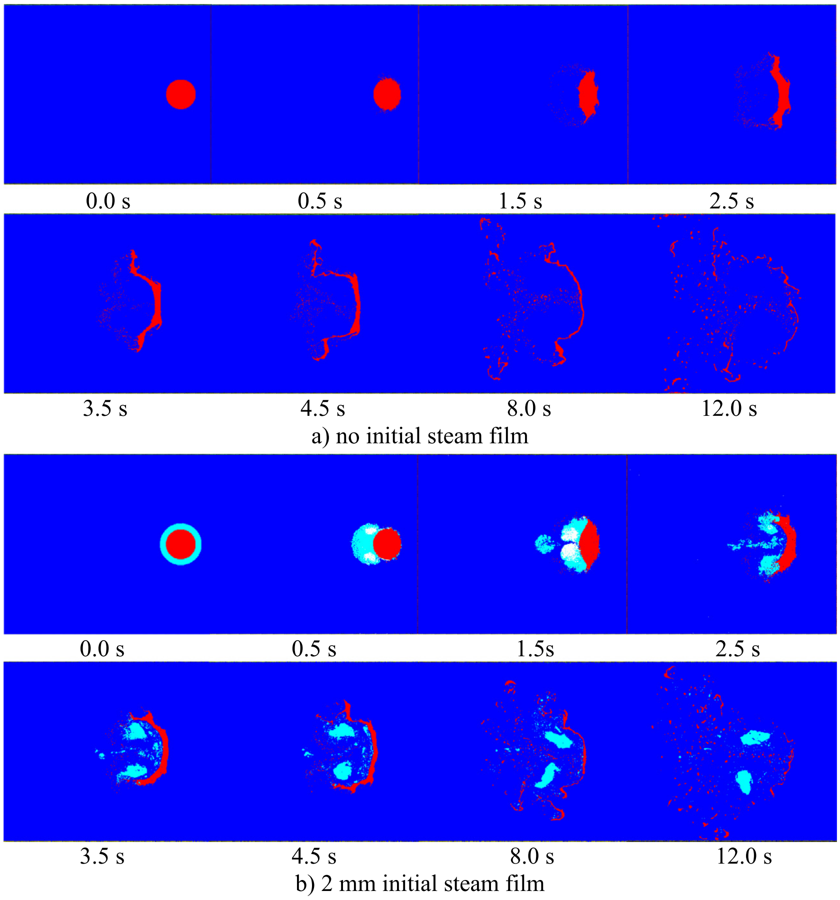


Fig. 14. Melt drop breakup behavior at We = 1818 [144].

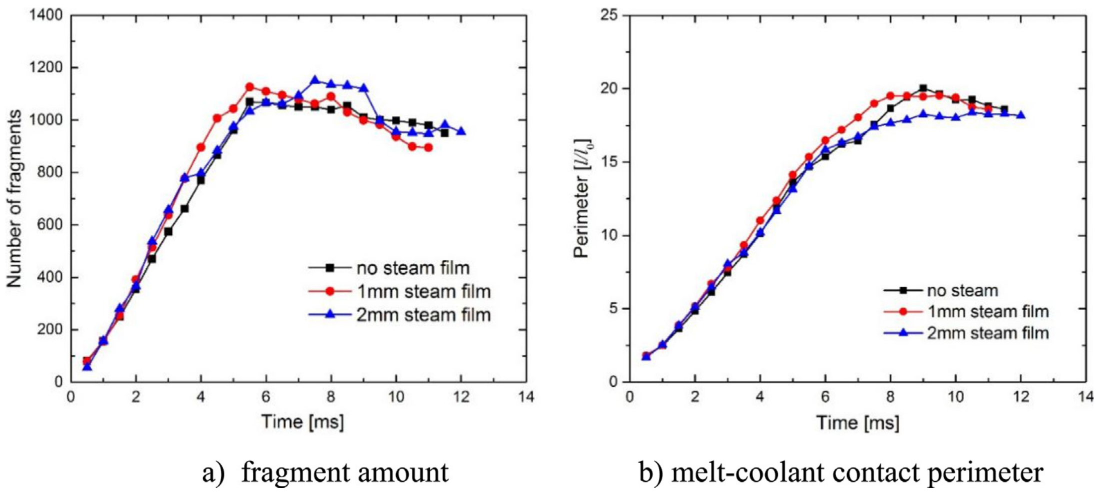
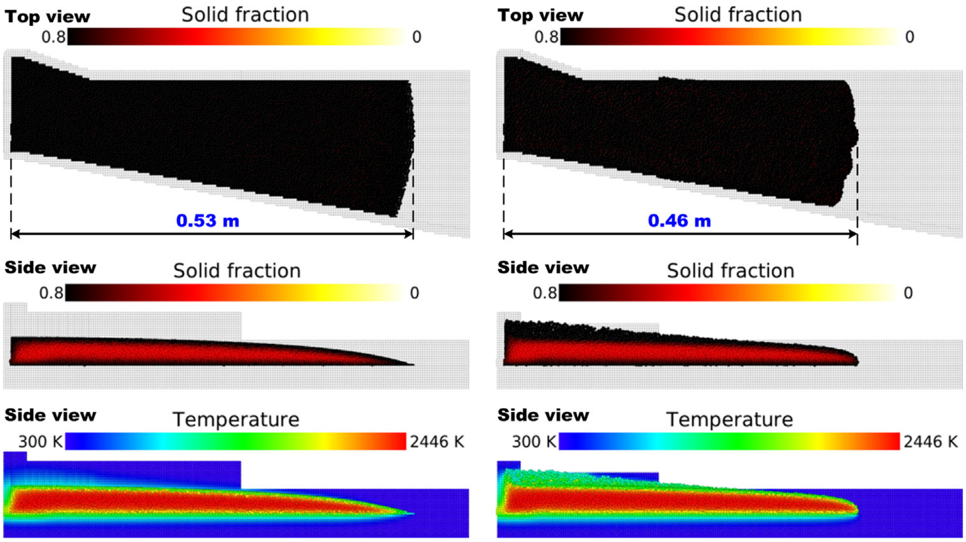


Fig. 15. Fragment amount and melt-coolant contact perimeter during melt drop breakup process at We = 2618 [144].

the effect of Zr oxidation on concrete ablation is investigated. The results indicate that Zr oxidation significantly increases ablation rate and the phenomena of crust formation and remelting are observed at the interface of melt



(a) Original algorithm (b) Duan's algorithm

Fig. 16. Distributions of solid fraction and temperature simulated by the original and Duan's algorithms [149].

and concrete. Crust can significantly slow down concrete ablation rate. Li and Yamaji [152] investigated isotropic and anisotropic ablations in MCCI by carrying out numerical simulations of CCI-2 and CCI-3 experiments. The interactions of fully oxidized PWR core melts with specially designed two-dimensional limestone and siliceous concrete are analyzed. The effects of slag film, crust formation, gas generation and aggregates on concrete ablation behavior are investigated. The obtained ablation profile and the axial and lateral ablation rates agree well with the experimental data. The results also prove that aggregates are the cause of anisotropic ablation profile in the interaction of melt with siliceous concrete. The representative ablation profiles in the simulation of CCI-3 experiment are shown in Fig. 17. Chai et al. [153] analyzed CCI-2 experiment as well by coupling multi-physics models with MPS method. Crust thicknesses at axial and radial directions are predicted. Chai et al. [154] also investigated the differences of concrete ablation profiles during melt interacting with limestone and silica concretes, respectively. Li et al. [155] investigated crust evolution in the interaction of molten corium jet with sacrificial material of the core catcher of European pressurized water reactor. The results indicate that crust and molten concrete layer present at UO_2-ZrO_2 melt and concrete interface, whereas no crust form in the interaction of Fe–Zr metallic melt with concrete. The crust experiences stabilization-fracture-reformation periodic process and the molten concrete layer accumulates and escapes simultaneously. The crust evolution processes are presented in Fig. 18.

On the other aspects, Mustari and Oka [156] and Mustari et al. [157] developed a molten corium eutectic reaction model based on the theory of mass diffusion. Phase diagram is employed to judge the state of materials. The developed model is applied to the analyses of eutectic reactions of U–Fe solid–liquid system and Pb–Sn solid–solid system, respectively. The analyzed penetration rates agree well with the experimental data. Li et al. [158] analyzed the dissolution process of uranium dioxide by molten zircaloy using mass diffusion model. The results indicate that the dissolution of UO_2 by molten zircaloy fits parabolic law, but the dissolution rate decreases after UO_2 reaches saturation in liquid. Natural convection of molten zircaloy has significant effect on dissolution process. Chen et al. [159] simulated melt penetration and solidification behavior in the instrument tube in RPV lower head, which is an important phenomenon affecting melt discharge. Melt flow resistance increases as the formation of melt crust. In all the analyzed cases, melt does not penetrate through tube due to the blockage of melt crust. Flow channels in fuel support piece of a BWR are potential corium relocation channels from core region to lower head. Melt solidification behavior in fuel support piece can affect melt relocation process and determine the formation of melt pool above core plate. The analysis by Chen et al. [160] points out that the channels in fuel support piece are plugged due to the solidification of zircaloy melt when initial temperature is low, and fuel support pieces are intact in all the calculated cases. Li et al. [161,162,163] investigated melt stratification and migration in debris bed in

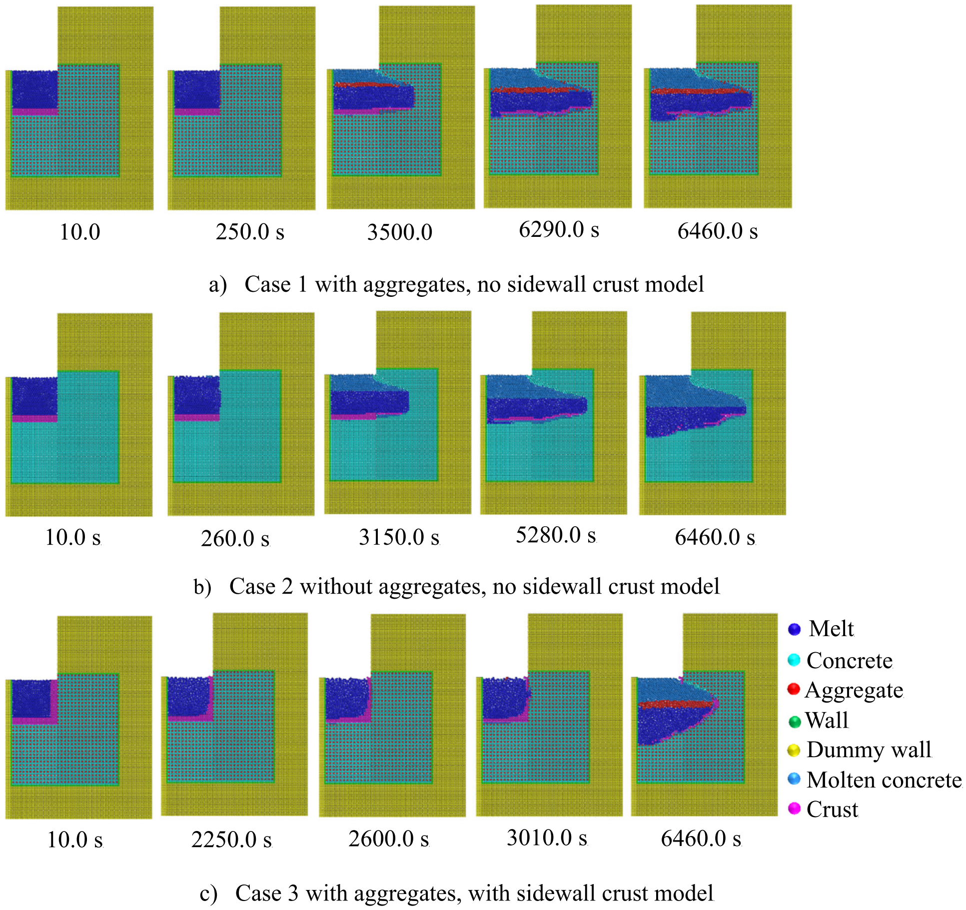


Fig. 17. Representative moments of simulated MCCI process in CCI-3 experiment [152].

RPV lower head, which have significant effects on melt heat transfer and heat flux distribution on vessel wall. Stratification behaviors of oxide and metallic core melt are investigated under different temperatures, viscosities, surface tensions, and boundary conditions. The results indicate that the crust formed at the oxide and metallic melt interface can impede stratification progress to a large extent. Melt penetration in debris bed is terminated by crust and penetration distance increases as the increase of debris bed porosity and the increases of melt and debris temperatures. Figs. 19 and 20 present melt stratification and migration in debris bed, respectively. Regarding to the thermal erosion of lower head wall by melt jet relocated from core region, Li et al. [164,165] investigated erosion rate of lower head wall impinged by separate melt components. The fastest erosion rate presents in different conditions that have different melt and superheat temperatures. Based on parametric studies, it is inferred that in the real severe accident the largest threat is from metallic melt jet, because thermally protective crust formed at the interface of oxide melt jet and vessel wall prevents erosion progression.

3.3. Brief summary

MPS method has proved its inherent advantages in nuclear engineering application compared with mesh-based method. Past studies have analyzed bubble dynamics in boiling and condensation processes and melt behaviors in nuclear reactor severe accident. However, in the simulations of bubble growth and deformation by MPS-MAFL method, the mass transfer between vapor and water phases is ignored, and as a result, mass conservation is not satisfied. MPS method is expected to be used to the simulations of violent boiling by considering mass, momentum

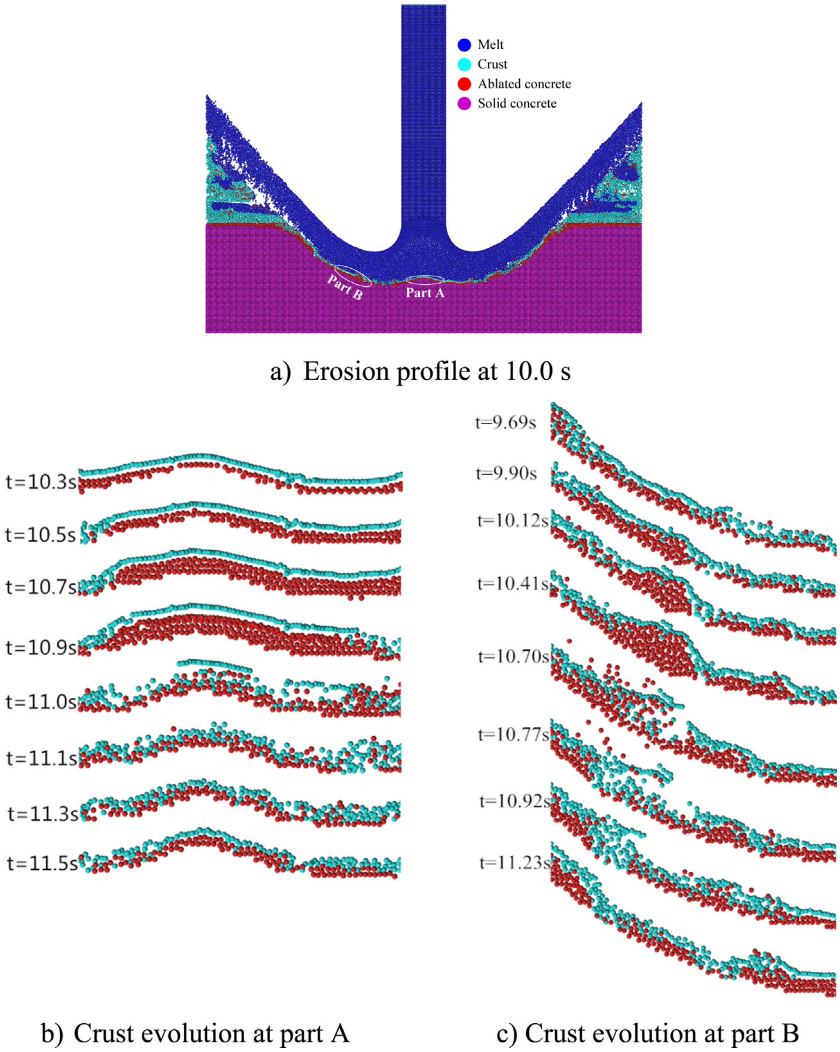


Fig. 18. Crust evolution processes in the interaction of UO_2-ZrO_2 jet with concrete, $T_j = 3000$ K and $D_j = 20$ mm [155].

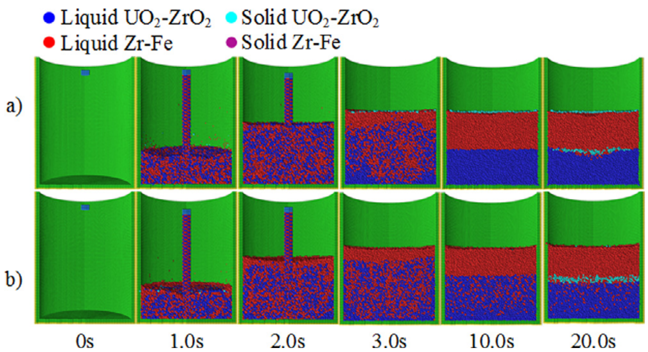


Fig. 19. Stratification processes of oxide and metal melt under different viscosities (initial melt temperatures in cases (a) and (b) are 2913 K and 2900 K, respectively) [163].

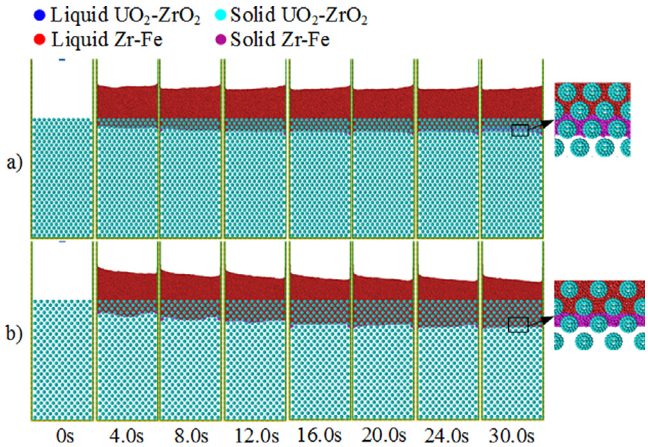


Fig. 20. Molten metallic corium migration in the oxide debris bed ((a) porosity = 0.435; (b) porosity = 0.552) [163].

and energy transfer and conservation between different phases. More models need to be developed and coupled together to deal with complex thermal hydraulic phenomena.

Molten corium has the characteristics of multicomponent and multiphase state, and its migration and phase transition processes are complicated, involving free surface, flow breakup and coalescence, decay heat release, etc. Past studies have proved MPS method to be a powerful tool for melt behavior analysis, which is irreplaceable by grid method. The models of fuel–coolant interaction, corium spreading, MCCI, melt stratification are established, and the mechanisms of these phenomena are investigated. At present, melt behavior analyses with MPS method mainly focus on the accident progressions in lower head and on ex-vessel concrete floor. Owing to the limitation of computation cost, the scales of computational domains of melt behavior analyses are small, and most of studies only compared simulations with the experimental results. Therefore, it is expected to conduct melt behavior analyses with the prototypical melt materials and under the conditions closing to real accident.

4. Future perspectives

The advantages of meshless particle method, characterized by easily capturing free surface, component tracking in multi-component flow and easily modeling solid–liquid phase transition, are complementary with the features of grid-based numerical method. MPS method has got substantial improvements in past two decades, and the application areas have expanded from initial nuclear engineering to ocean, mechanical, and chemical engineering. Even though prominent advancements have been achieved in the aspects of stability, accuracy, boundary conditions, multiphase flow, and fluid–structure interaction, there are still many challenges in the practical applications of engineering problems. Therefore, further fundamental and comprehensive improvements are necessary to make MPS method be a reliable and robust numerical tool in engineering analysis.

A critical defect of MPS method is its low accuracy and instability, which are about one order of magnitude lower than that of grid method. Exactly speaking, the standard MPS method only has zero-order of accuracy. Past studies have tried various approaches to improve its accuracy and stability, including artificial inter-particle force, error-compensating parts in PPE source term, PS scheme, and corrective matrixes for particle interaction models. Some improvements in accuracy and stability are gained with these treatments, but the conversation and convergence of numerical schemes are not strictly preserved, because some of the models are not derived mathematically from fluid mechanics. A potential method to thoroughly resolve the low accuracy and instability problems of MPS method may be the least squares MPS method, which revisits fluid mechanics and Taylor series expansion. It does not need any artificial pressure stabilizer. Preliminary studies by Tamai and Koshizuka [39] and Matsunaga et al. [41] indicate that the stability and global high-order accuracy and consistency of least squares MPS method are demonstrated. In future, therefore, the studies should be focused on the derivation of flow governing equations in a mathematical approach to preserve the conservation and convergence of numerical scheme. Like least squares MPS method, the calibration coefficient should be avoided. Moreover, numerical schemes should be robust in the simulation of complex flows not only in the limited conditions.

Another challenge of particle method is the modeling of multiphase flow with a large density ratio at phase interface. Various approaches have been tested including calculating two phases separately, density smoothing, and particle–grid hybrid method for gas–liquid flow. However, some of these treatments violate mass and momentum conservation laws, and the numerical schemes are not consistent at interfaces. In the future, research topics should focus on the conservations of volume, momentum and energy at interface and the sharpness of interface to preserve original physical properties of materials. Considering the applications to solve multiphase flows in nuclear engineering, phase transition model is expected to be developed for evaporation and condensation simulations. In regard of solid–liquid flow, the MPS-DEM coupling method is a promising method to calculate the interactions of solid–liquid and solid–solid. To make it more robust, stability enhancement techniques should be applied to the simulations of two-phase flow, because the spurious hydraulic force caused by pressure fluctuation may lead to inaccurate motion of solid particulates. Moreover, the simulation of irregular solid particulates should be considered as well.

Development of turbulence models for particle method are urgent needed, because turbulence flow is a ubiquitous phenomenon in nuclear and other relevant engineering. Several studies incorporate large eddy simulation model to SPH and MPS particle methods for the simulation of low Reynolds turbulence flows [166–168]. When this model is applied to high intensity turbulence, the flow field cannot be predicted accurately, especially in the conditions with heat transfer. The concepts of time-averaged and spatially averaged turbulence models that have been widely applied in mesh-based method can serve as references to develop robust turbulence models for particle method. Moreover, heat transfer enhancement by turbulent vortexes should be taken into account in the solving of energy conservation equations.

In regards to the applications in nuclear engineering, the simulations of some fundamental thermal hydraulic phenomena and melt behaviors have been conducted, but these simulations are simple and small scale. It is expected to investigate thermal hydraulics considering more real reactor conditions, such as vaporization and condensation, and to analyze melt behavior using real melt properties and boundaries. Moreover, the capability of MPS method to conduct large-scale simulation is necessary and important. This capability should consider the reduction of MPS computation cost and the convenience to build particle configuration of complex geometry. On the other aspect, there is a gap between the current MPS technologies and the actual application; that is, the advanced techniques such as stability and accuracy improvements and multi-resolution technique are not fully used in nuclear engineering. In future, therefore, it is important to develop MPS method considering the application background, and introduce advanced techniques to nuclear engineering application.

5. Concluding remarks

The developments of MPS method and its applications in nuclear reactor thermal hydraulics are reviewed. The state-of-the-art advancements in stability and accuracy, boundary conditions, surface tension, multiphase flow, fluid–structure interaction and multi-resolution technique are discussed. Applications in nuclear reactor fundamental thermal hydraulics and melt behaviors in severe accident are summarized. The drawbacks, current challenges, future developments and potential application areas of MPS method are highlighted. To conclude, MPS method is at an exciting stage in its development, but the consistency, conservation and convergence in physics and mathematics should be preserved during model developments. New application areas of MPS are essential in helping to push method development forward.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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